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Firm Adaptation in Production Networks: Evidence from Extreme Weather Events in Pakistan[†]

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This paper considers how far private adaptation may reduce future vulnerability to climate risks. Using data on monthly firm-to-firm transactions from Pakistan, we find that flood-affected firms are more likely to relocate to safer ground and shift purchases toward suppliers in less flood-prone regions and reached via less flood-prone roads. The results indicate that firms are imperfectly informed about flood risk and update their beliefs following floods. We quantify aggregate impacts using a spatial model of endogenous production network formation. The findings suggest that firms' learning will shape the economic impact of increasingly frequent climate-related disasters. (*JEL* D22, L14, O13, O14, Q54, R32)

Climate change presents a global threat to human populations and economic growth. Despite growing policy and research focus on mitigating climate risks, it is now clear that mitigation efforts will be insufficient to prevent many of their damaging effects. Foremost among these is the increased likelihood and severity of extreme weather events (IPCC 2021). Estimating the costs of climate change, and designing appropriate policies to moderate damages, requires an understanding of how those affected by climate disasters respond to these changing circumstances (Carleton et al. 2022; Bilal and Rossi-Hansberg 2023). In particular, in developing economies, where capacity for centralized policymaking is often weak (Greenstone and Jack 2015), the burden of adaptation often lies disproportionately with private actors. This paper considers how firms—the primary sites of economic activity and central to population welfare—anticipate and adapt to climate-related shocks.

Estimating firm adaptation to climate change is challenging because both risk exposure and adaptation margins may involve complex network effects. Firms are exposed to spatially concentrated disaster risk not only directly because of their production taking place in risky locations, but also indirectly via exposure of their

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suppliers or buyers (Barrot and Sauvagnat 2016; Boehm, Flaaen, and Pandalai-Nayar 2019; Carvalho, Nirei, Saito, and Tahbaz-Salehi 2021), or transportation infrastructure that links firms to their trading partners (Korovkin, Makarin, and Miyauchi 2026). Firm responses to such disasters can therefore reduce their risk exposure along several margins: Firms may exit or contract activities in risky locations, relocate toward less disaster-prone regions, adjust their mix of trading partners, or shift routes toward those less exposed to disaster risk.[1] Capturing exposure and adaptation margins therefore requires detailed knowledge of production linkages, as well as a convincing methodology to distinguish changes in expectations over supply partners' outcomes from changes in costs or other determinants of supply chain formation.

Our empirical analysis leverages detailed data on transactions between firms, as well as measures of both firm- and supply route-level exposure to natural disasters, to provide evidence that firms undertake adaptive production and sourcing decisions in the aftermath of major floods. The context of our study is Pakistan, one of the countries most exposed to extreme weather worldwide (Eckstein, Künzel, and Schäfer 2021). We study firm and production network adaptation from 2011 to 2018 at a highly granular spatial and temporal scale using a series of novel datasets. We leverage georeferenced monthly microdata on the near universe of formal firm-to-firm sales transactions to capture the key adaptation margins available to firms at a high frequency and level of precision. The transaction records are complemented with over six billion observations from GPS trackers installed on more than 15,000 commercial trucks over the same period to measure the extent to which supply routes are affected by natural disasters. Flood disruptions to firm activities and the road network are measured by intersecting these with satellite-derived data on major flood events. To capture how far responses to these events may reduce vulnerability to future floods, we supplement these data with high-resolution measures of flood risk derived from a global flood hazard model.

We first document severe but short-lived disruption of firm activities and traffic induced by flooding of firm premises and roads. Sales and purchases of severely flooded firms (defined as those for which more than 10 percent of land within a 2-kilometer (km) radius is flooded) decline by 9.7 percent and 3.9 percent, respectively, in the month of recorded flooding, though both recover within six months. Impacts are intuitively strongest for more severely flooded firms and for manufacturing, retail, and wholesale firms relative to services. Following severe flood events, the probability of firm exit increases. Flooding of roads also leads to large but brief disruptions to traffic flows: Mean truck speeds decline by 1 km/hour and truck-day counts by 14–17 percent immediately following floods, with reversion of both outcomes to preexposure levels within a month. The core of our analysis then turns to the question of whether these significant but temporary flooding disruptions induce firms to undertake longer-term adaptive changes that reduce their vulnerability to future flooding. We consider evidence for adaptive adjustments along the margins of firm relocation, supplier choice, and supply route choice.

[1] IPCC (2022) defines adaptation as "adjustment to actual or expected climate and its effects in order to moderate harm or take advantage of beneficial opportunities." As discussed in more detail below, this may include both forward-looking decisions intended to reduce future vulnerability, and reductions in risk that result from the direct impacts of disasters. Both may be "adaptive" by this definition, but given their potentially divergent policy implications, we present evidence that aims to distinguish the two channels in our analysis.

We provide the first micro-level evidence of firm-level adaptive relocation by studying whether flooded firms relocate toward areas less prone to flooding. The results suggest that severely flooded firms see a 3.67 percentage point increase in the probability of relocating more than 10 km away over the study period relative to those that are not flooded, a large increase relative to the 17.34 percent probability of moving by more than 10 km among nonflooded firms. Importantly, this relocation is adaptive in the sense that flooding induces firms to relocate systematically toward less flood-prone locations: Severely flooded firms that relocate more than 10 km see a 20 centimeter (cm) reduction in the expected flood depth that they would experience during a 1-in-100-year flood. District-level gravity specifications also suggest that relocating firms respond to recent flooding in deciding on a destination location, substantively avoiding locations that have recently been flooded within origin-destination district pair moves. These firm-level findings complement a recent climate migration literature that considers the response to extreme weather events of populations (Boustan, Kahn, and Rhode 2012; Mueller, Gray, and Kosec 2014), night lights (Kocornik-Mina, McDermott, Michaels, and Rauch 2020; Elliott, Strobl, and Sun 2015), and employment (Indaco, Ortega, and Taşpınar 2021).

Given that firms may be exposed to climate risk not only directly but also via vertical linkages, we use transaction-level data to examine adaptation through supplier choice. Diversification may ameliorate expected flood losses by reducing dependence on individual suppliers, and spreading risk across suppliers with uncorrelated shocks (Cole, Elliott, Okubo, and Strobl 2019; Meltzer, Ellen, and Li 2021; Boehm and Sonntag 2023; Castro-Vincenzi, Khanna, Morales, and Pandalai-Nayar 2024). Consistent with this, we find that firms increase the number of suppliers from which they source following flooding of their suppliers, but this response endures for less than a year. Combining transaction-level data with data on flood risk reveals that these firms more persistently shift the *composition* of their supplier base toward less flood-prone suppliers. These adaptive adjustments to supplier choice are intuitively especially pronounced among smaller firms (whose simpler supply networks may be less costly to adjust), single-branch firms (with fewer opportunities to hedge risk across diversified production sites), and firms in industries that require more relationship-specific inputs (for which flooding of upstream suppliers may be more costly). Importantly, this adaptive behavior is also evident among a firm's nonflooded suppliers—suggesting a role for forward-looking adaptation rather than simply the mechanical effect of no longer being able to source from flood-affected sellers—and persists for at least four years after flood exposure.[2] These results highlight that accounting for network-based adaptation margins is important and demonstrate the sophisticated nature of firms' adaptive responses beyond the direct flood exposure of production sites.

The vulnerability of the firm network is predicated not only on the flood risk of firms, but also on the riskiness of the trading links connecting them. We use data on flood-induced road disruptions to examine adaptation via firms' choice of supply routes. Route-level specifications leverage the bilateral nature of the transaction-level

[2]This is consistent with extensive margin evidence from financial data that temperature and flood shocks at supplier locations that exceed expectations may induce customers to terminate relationships and choose replacement suppliers with lower expected climate risk (Pankratz and Schiller 2024).

data to fully isolate adaptive behavior by using buyer-seller, buyer-time, and seller-time fixed effects. These control for any direct effects of floods on firms and rule out potentially confounding shocks that may affect flooded firms even after their sales and purchases have recovered, such as local labor market disruptions or correlated cost shocks. The results suggest that firms respond to short-lived flood-induced disruptions to road transportation by reducing their dependence on supply partners reached via flood-prone routes. Despite preflood traffic flows being restored within a month, firms do not switch back to sourcing from these suppliers once access is restored. This provides our most cleanly identified evidence of firms undertaking long-term adaptation in response to transient shocks and highlights the importance of accounting for route- as well as firm-level adaptation margins.

Taken together, these results demonstrate that firms respond adaptively to climate shocks and may therefore play an important role alongside governments and households in adaptation to climate risks (Grover and Kahn 2024; Goicoechea and Lang 2025). As discussed in Carleton, Duflo, Jack, and Zappalà (2024), climate adaptation may occur both ex ante ("anticipatory") and ex post ("reactive"); our evidence is consistent with a key role for reactive adaptation in response to flood events in this context. This may encompass both forward-looking actions intended to reduce future vulnerability and more "mechanical" effects of destruction: For instance, a firm that exits following a flood may do so in expectation of lower profitability in the future, but this may also simply reflect capital destruction experienced during the flood. Given that both sources of adjustments will alter future vulnerability of the production network, we aim to capture both in our analysis of adaptation. While several of the adaptation margins we identify might plausibly reflect a combination of both effects, the finding that firms shift toward suppliers that are less flood prone even among their nonflooded suppliers highlights an important role for forward-looking adjustments. This has crucial implications for our understanding of the role of climate risk in firm decision-making, complementing a recent literature examining individual decision-making and learning in relation to climate risks (Deryugina 2013; Kala 2017; Patel 2024). More broadly, the complex adaptive behavior we identify is informative about the forward-looking behavior of firms in the presence of risk, which plays a central role in a range of economic and policy questions.

While a worsening trajectory of natural disaster risk will have damaging impacts for firms, our results suggest that these may be moderated by firms responding adaptively as more information becomes available. Quantifying the economy-wide implications of this adaptation for the vulnerability of the production network is challenging given that aggregate effects will reflect firm connections via multistep linkages and general equilibrium forces. We therefore develop a quantitative spatial model of endogenous production network formation that captures firm-to-firm linkages and general equilibrium effects in order to estimate aggregate impacts.

The model features firms that are subject to both idiosyncratic and aggregate flood shocks that reduce firm productivity. Firms are imperfectly informed about the distribution of these risks but update their beliefs in response to flood shocks. The framework builds on recent advances in modeling production network formation under uncertainty (Kopytov, Mishra, Nimark, and Taschereau-Dumouchel 2024) to incorporate this imperfect information. Before flood shocks are realized, firms search

for suppliers in different locations, taking into account their beliefs over potential partners' flood risk. These search decisions affect the distribution of supplier draws that the firm receives. Once shocks have been realized, firms make production and sourcing decisions conditional on these draws to minimize costs. We incorporate insights from the spatial trade literature, leveraging extreme value distributions to yield tractable gravity equations describing sourcing shares (following Eaton and Kortum 2002; Oberfield 2018; and Boehm and Oberfield 2020, 2022). These gravity equations allow us to estimate flood-induced productivity shocks and identify the role that adaptive changes in firms' supplier search decisions play in shaping the economy's exposure to floods.

We parameterize flood-induced productivity shocks at the level of locations comprising proximate firms with similar supplier flood exposure, which update their beliefs in a similar way following flood events. Accounting for direct productivity losses as well as general equilibrium impacts of supplier disruption, the estimated economy-wide increase in the household cost index for the floods in our sample ranges from 0.05 percent to 0.3 percent. We use the model to estimate the impacts of adaptation undertaken in the aftermath of these floods via counterfactual simulations that assess how flood damages differ if we shut down adaptation following previous floods. This exercise reveals that adaptation following the 2012 floods helped reduce damages from subsequent floods affecting similar locations in 2013 and 2015, which would have been 5 percent and 1 percent higher, respectively, under sourcing shares that prevailed before the 2012 floods. However, adaptation in the aftermath of the 2012 floods is estimated to have *worsened* the impacts of the 2014 floods, which affected spatially disjoint regions in lower flood risk areas. This highlights that adaptive postflood responses, while reducing firm exposure to flood *risk*, need not always ameliorate damages from individual *realized* future flood events. This is especially true when realized flooding affects areas that are not particularly flood prone, which may become increasingly pertinent as flooding incidence responds to climate change.

Our results also suggest that there is significant scope for further adaptation in our setting. A large majority of firms have access to alternative suppliers that are nearby, in the same industry, and subject to lower flood risk than their flooded suppliers. Using our estimated model, we also find that the majority of locations would be able to reduce the variance of cost increases induced by flooding without increasing average costs, by changing their supplier mix to source relatively more from less flood-prone locations. These findings point to the potential for policies that overcome barriers to adaptation to help firms to realize the benefits available from cost-effective improvements in resilience.

The paper's findings suggest that natural disaster risk plays an important role in firm decision-making and that the realization of climate shocks influences firm expectations in a meaningful way. This leads to adaptive adjustments along location, supplier and route choice margins, with complex system-wide effects as a result of interlinkages in production networks. A significantly worsening trajectory of flood events is predicted in Pakistan (World Bank and ADB 2021) and globally (Kirezci et al. 2020) over the coming decades as climate change unfolds. As such, these responses will have profound implications for how firms will adapt to an increasingly risky environment, which will in turn be a key factor in determining the costs

of climate change. Our results indicate that firms are imperfectly informed about climate-related disaster risk, highlighting the potential for policies addressing such frictions to ameliorate climate damages.

The remainder of the paper proceeds as follows. Section I describes the empirical setting and datasets used in the analysis. Section II provides evidence for the disruptive impacts of flood events on firm production and road transportation in Pakistan. Section III examines firm and supply chain adaptation in the aftermath of flood events. Section IV uses a quantitative model of production network formation and adaptation under supply chain uncertainty in our empirical setting to understand the importance of adaptive decisions for aggregate outcomes. Section V concludes.

I. Setting and Data

The empirical setting for our analysis is Pakistan, one of the world's most vulnerable countries to the effects of extreme weather (Eckstein, Künzel, and Schäfer 2021; Guha-Sapir, Below, and Hoyois 2022), where rapid industrialization is proceeding alongside increasing vulnerability to the effects of climate change. Floods are preeminent among these and result in severe disruption to firms and infrastructure. The 2022 floods alone are estimated to have resulted in damages of $40 billion (PMO 2022) (roughly 11.5 percent of 2021 GDP), 30 percent of which are accounted for by damages to infrastructure and nonresidential structures (World Bank 2022).

Transportation infrastructure is especially affected by flooding: The 2022 floods damaged over 8,000 miles of roads and 392 bridges (Congressional Research Service 2022), while the 2010 floods are reported to have damaged 10 percent of the country's road network (World Bank 2010).

In this context, the analysis draws on four novel georeferenced micro-datasets from 2011 to 2018 in order to characterize flood-induced disruption to firms and production network linkages, and identify adaptive adjustments at a fine temporal and spatial resolution. Firm-to-firm transaction data allow us to identify production network linkages, disruption to firms and relationships, and examine adaptation via firm location and supplier choice. We use GPS tracker signals from commercial trucks to estimate disruption to transportation routes and examine adaptation via trading-route choice. We identify the flood exposure of firms and roads in the data by intersecting these geocoded datasets with satellite images of flood extents. Finally, detailed data on flood risk from an advanced flood hazard model help us to characterize adjustments as adaptive to the extent that they reduce the flood risk of firms' premises and supply network dependencies. We describe key features of each dataset here; further details on data construction are provided in Supplemental Appendix C.

A. *Firm Transactions Data*

Data on firm outcomes come from the near universe of formal firm-to-firm monthly sales transactions for all VAT-registered firms over July 2011–June 2018 from Pakistan's Federal Board of Revenue (FBR 2019). At the firm level, these data contain information on reporting firms' name, industry, and address at the beginning

and end of the study period. The data also contain monthly information on all transactions where at least one party is registered for VAT, as well as total reported firm sales, purchases, exports, and imports. We construct three firm-level monthly sales and purchases variables: one given by the firm's reported sales (purchases); a second given by the sum of transaction-level sales (purchases) reported by the firm; and a third, which aggregates the union of transaction-level purchases (sales) reported by the firm and its trading partners.

The data contain reports of all firms in Pakistan registered to pay VAT, which is required for all importers, wholesalers, and distributors, as well as manufacturers and retailers with revenue exceeding 10 million rupees in the previous tax-reporting period and an annual utility bill above 800,000 rupees.[3] This yields a raw dataset containing information on 419,517 firms, which either self-report or are reported upon in the reports of VAT-registered firms.

We take a number of steps to exclude incomplete or potentially misreported transaction data. We exclude firms that have been identified as—or transact exclusively with—"invoice mills," firms that exploit breaks in the supply chain to purchase and sell VAT invoices without conducting any real business (Waseem 2019; Keen and Smith 2006). This removes 4 percent of firms in the sample. Also excluded are 29 percent of firms for which there is insufficient address information to geocode the firm's location in 2011 and therefore for which we cannot identify flood exposure. A large fraction of the remaining firms report very infrequently or not at all (the latter appearing in the dataset only by virtue of having transactions reported upon by their VAT-registered transaction partners). Given that measurement for such firms is likely to be poor and that singleton observations will not be informative for studying the effects of flooding, we also exclude firms that report at most twice in any transaction measure. The full set of sample restrictions reduces the firm count considerably to 73,339, but excludes firms that account for only 2.9 percent of aggregate sales and 3.4 percent of aggregate purchases.

The resulting data represent a large fraction of the economic aggregates reported in national accounts (Pakistan Bureau of Statistics 2025). In the restricted sample, aggregate manufacturing value added accounts for 89 percent of reported manufacturing GDP in the last year of the sample. Total value added in our restricted sample accounts for approximately 20 percent of total GDP, since the reported aggregates include sectors that are generally not subject to VAT—agriculture, certain services, and the informal sector. Industry and product codes are not available for all firms in the sample, but among those firms for which a sector can be identified, 37 percent are classified as manufacturing firms, 29 percent retail/ wholesale, 20 percent services, 9 percent agriculture, and 4 percent other.[4]

[3] These thresholds were raised from five million rupees in July 2016 and from 700,000 rupees in July 2015, respectively.

[4] Firms and products in these data are classified using the Pakistan Standard Industrial Classification (Federal Bureau of Statistics 2010) and the Pakistan Customs Tariff codes (Federal Board of Revenue of Pakistan 2013), with capital goods identified from the Fifth Schedule to the Customs Act (Federal Board of Revenue of Pakistan 2015). Industry codes are supplemented, for firms with valid national tax numbers, using Orbis (Bureau van Dijk 2023). The heterogeneity analysis additionally draws on the Census of Manufacturing Industries (Federal Bureau of Statistics 2006) and on measures of input relationship specificity (Boehm, Oberfield, South, and Waseem 2025a) constructed for Boehm, Oberfield, South, and Waseem (2025b).

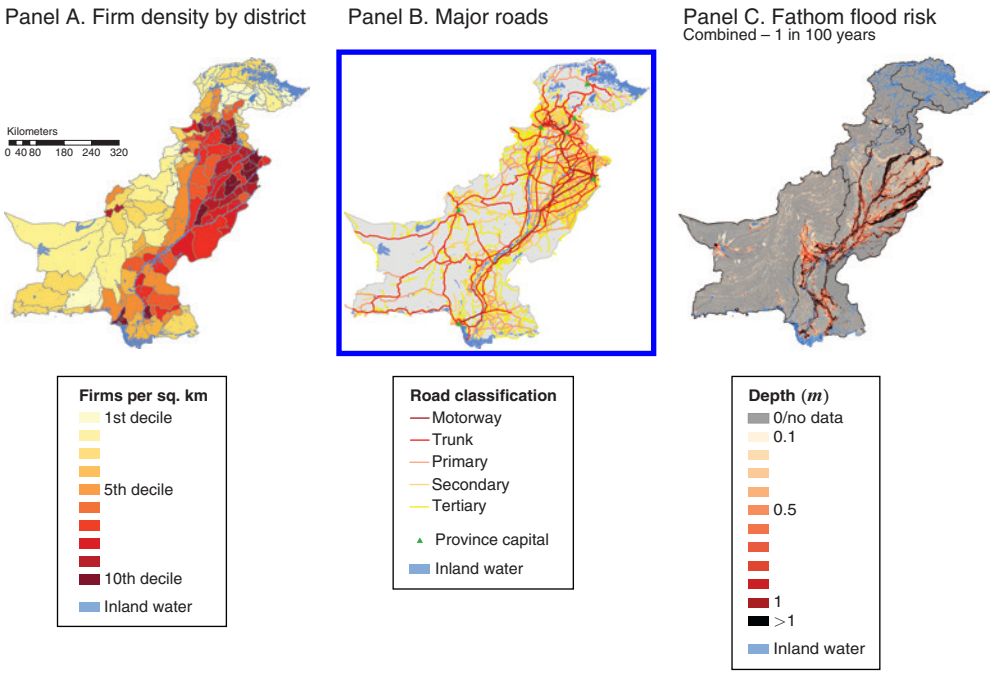


FIGURE 1. SPATIAL DISTRIBUTION OF FIRMS, ROADS, AND FLOOD RISK

Note: Flood risk in panel C is defined as the maximum across fluvial and pluvial flood risk, both measured as the expected flood depth in meters for a 1-in-100-year return period.

To capture entry and exit of firms while allowing for the potentially confounding effect of irregular reporting, we define a firm as entering on the date of their first report (self-reported or reported by a transaction partner) if this is more than a year since the beginning of our panel, and as exiting on the date of their last report if this is more than a year from the end of our panel. All observations for a given firm before their date of entry or after their date of exit are set to missing. Summary statistics describing the firms and transactions in the restricted sample are included in Supplemental Appendix Tables A.1 and A.2, respectively.

Address information for firms in the sample was used to geocode firm locations using the Google Maps API (Google 2021).[5] The location of firms in the sample is shown in panel A of Figure 1 and displays a strong concentration of firms in Pakistan's major industrial provinces of Punjab and Sindh. Seventy-two percent of firms, accounting for 74 percent of sales, are in urban areas (Esri 2025), predominantly concentrated in the major cities of Karachi and Lahore. For 60 percent of firms,

[5] For those firms for which no address information was available from the FBR's firm transactions data, where possible, we used address information scraped from the FBR's Active Taxpayer Lookup Portal. Where multiple addresses are available for a firm, we use the primary "business" address. We drop a small number of firms reporting two business addresses that are more than 5 km apart. Geocoded firm locations are assigned to administrative units using the GADM database of global administrative areas (Global Administrative Areas 2018) and the OCHA subnational administrative boundaries for Pakistan (OCHA Field Information Services Section 2018).

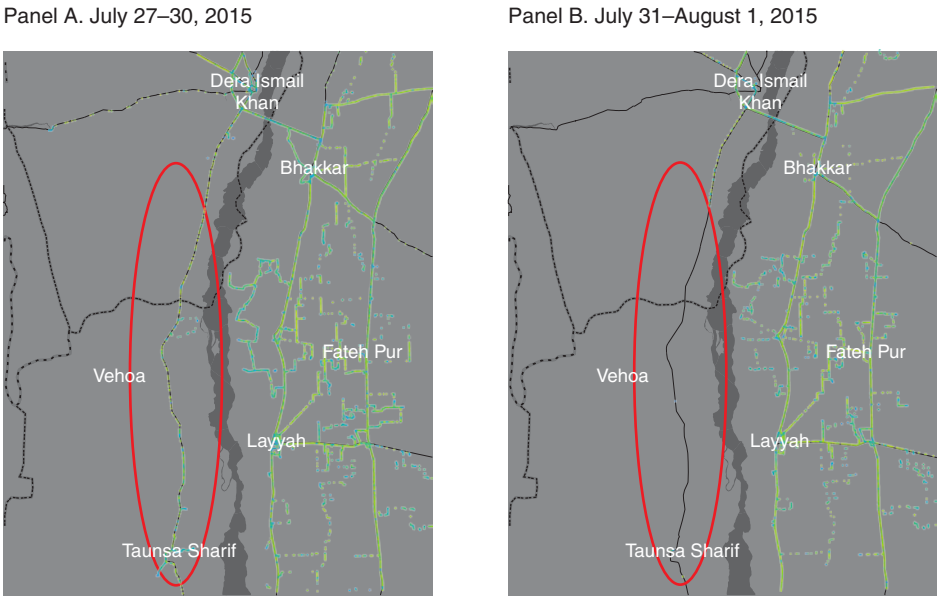


FIGURE 2. N-55 INDUS HIGHWAY FLOODING DISRUPTION

Note: The maps display the location and speed (denoted by color gradient in km/hour) of trucks from GPS tracker data in the area surrounding the N-55 highway near Vehoa in Punjab Province before (panel A) and after (panel B) reported flooding.

sufficient address information is available to geocode addresses in 2011 and 2019 separately.

B. *GPS Tracker Data from Commercial Trucks*

We obtain high-frequency data from GPS trackers installed in more than 15,000 commercial trucks in Pakistan from a large original equipment manufacturer (Tracking World Pvt. Limited 2020). The data provider sells tracking devices and associated tracking and fleet management solutions to truck manufacturers, logistics providers, industrial and insurance companies. The data comprise more than six billion observations showing the precise location, time stamp, and speed of trucks traveling on Pakistan's road network from 2012 to 2018.

Figure 2 displays the capacity of this data to capture flood-induced disruption to roads at an extremely fine spatial and temporal resolution. The figure shows the area surrounding the N-55 highway near Vehoa in Punjab Province (circled in red), which at 09:15 on July 31, 2015, was reported by Pakistan's National Disaster Management Authority (2015) to have been hit by floodwater, which "swept away a 300-foot portion of the highway" (Dawn.com 2015). The left-hand panel shows normal traffic, reflected in continuous GPS signals, running along the encircled north-south highway in the four days leading up to the flood. The right-hand panel shows the abrupt cessation of traffic along the route in the direct aftermath of the flooding from 10:00 on July 31 to 23:59 on August 1. In Section IIB, we use weekly

road edge-level regressions to document a systematic pattern of such flood-induced disruptions to road traffic in our sample.

We construct firm-pair-route-level measures of travel speeds and disruption over time. To do so, we obtain OpenStreetMap data on Pakistan's road network comprising motorways, trunk roads, primary, secondary, and tertiary roads and their links (Geofabrik GmbH 2021), shown in Figure 1, panel B. These are split at road endpoints and intersections to yield an edge-level dataset onto which we project the GPS tracker observations using the closest edge within ten meters of the observation coordinates.[6] Consecutive observations are filtered out where the between-observation elapsed time is more than 30 minutes (periods during which the truck is likely parked) or the Euclidean distance is more than 20 km (from which sensible route information cannot be inferred). Using the remaining data observations, we find the shortest distance between each consecutive pair of observations along the edge network and—based on the observations' time stamps at both points—infer the average speed at which the truck traveled on all edges between them. This method overcomes selection bias in the speeds reported by the trackers themselves, arising from the fact that GPS trackers are disproportionately likely to report when vehicles are starting, stopping, braking, or turning. We aggregate speeds first to the day-truck-edge level, and then by taking the mean to the week-edge level, also calculating the number of truck-day observations within each week-edge ("day-truck count"). Supplemental Appendix Figure A.1 compares calculated speeds for an area of Lahore in 2015 to those reported for 2010 in Japan International Cooperation Agency (2012), and finds a high degree of overlap in both the magnitude and spatial distribution of reported speeds.

The edge-week-level data are used to construct the least-time route and travel time between each buyer-seller pair on average across nonflooded weeks, and during each week when flood events are recorded. Buyer and seller firm locations are projected onto the road network and the least-time route between them calculated using average edge-level speeds over the relevant period, weighting by edge length.

C. *Flooding Data*

Data on flood events in Pakistan from 2011 to 2018 are obtained from the United Nations Satellite Centre (UNOSAT) flood portal (United Nations Satellite Centre 2011–2015). This service provides satellite imagery of major flood events in response to requests from organizations such as UN entities, member states, government offices, and NGOs, most often to aid disaster response efforts.[7] These images allow us to map the exact location of floods, from which we extract a reference water layer comprising rivers, lakes, and other existing bodies of water (Geofabrik GmbH 2021).

[6] All observations with coordinates more than ten meters from any road edge in our data are discarded.

[7] We cross-reference the floods identified from this source with major flood events identified in other key natural disaster datasets (the EM-DAT dataset of the Centre for Research on the Epidemiology of Disasters, the Dartmouth Flood Observatory, and Sentinel Asia) to confirm that major flood events described in these sources are captured in our data. Relative to these sources, the UNOSAT data provide the advantage of exact flood locations and extents as observed from satellites during our study period. Matching the floods in our data with those in the EM-DAT dataset by province and date allows us to classify floods into riverine flood, flash flood, and general flood categories; results are not statistically significantly different across these categories.

For firm-level specifications, we aggregate satellite images to the monthly level, which yields a total of seven monthly flood events over 2011–2018. The aggregate extent of flooding during years in which we observe flood events during our sample is shown in Supplemental Appendix Figure A.2. We capture flooding of firm locations using the maximum share of a 2 km radius circular buffer surrounding the firm's geocoded location that is flooded during a given flood event. Using a small buffer around a firm's geocoded location reflects the fact that there can be local imprecision in firm geocodes and that local flooding may affect firms through, for instance, local labor market impacts as well as premises or capital destruction. As described in Supplemental Appendix Table A.1, using this definition, 30 percent of firms in the sample are ever flooded during the sample period, and 5 percent are flooded more than once. Nineteen percent of firms ever experience flooding of a supplier, where we define supplier firms as those that accounted for more than 10 percent of their purchases over the preceding three months. In a given flood, flooding of a firm's own premises and that of its suppliers are not extremely closely correlated, allowing us to distinguish the impacts of these two types of exposure separately: For a firm flooded (not flooded) in a given flood event, the probability that at least one of their suppliers is flooded is 23 percent (2 percent). Supplemental Appendix F.6 shows that results are qualitatively robust to choosing alternative buffer radii of 1 km and 3 km.

Given that roads are often disrupted by floods for shorter durations, and the extremely fine temporal resolution of the GPS tracker data, we consider flood-induced road disruption at the weekly level. Satellite images grouped at the weekly level yield a total of 11 flooded weeks during the sample period for which we observe GPS network data (2012–2018). Road network edges are intersected with the union of flood polygons observed in each week to identify the share of each road edge that is flooded. At the buyer-seller level, 46 percent of ordinary-time shortest routes experience flooding at least once during the sample period.

D. *Flood Risk Data*

Data on flood risk come from Fathom-Global (Fathom 2019), which uses a global flood hazard model combined with detailed terrain and hydrography data. The resulting datasets capture fluvial and pluvial flood risk (measured as the expected flood depth in meters) with return periods of 1 in 10 years, 1 in 50 years and 1 in 100 years, at a resolution of 90 meters.[8] For each return period, we take the maximum of the projected fluvial and pluvial flood risk. Panel C of Figure 1 maps the Fathom flood risk across Pakistan for a return period of 1 in 100 years; equivalent maps for the other return periods are shown in Supplemental Appendix Figure A.3. These maps demonstrate a significant degree of overlap between flood-prone locations and areas with a high density of firms and roads, shown in panels A and B of Figure 1.

The flood risk of a firm location is calculated as the average Fathom flood risk depth index in the 2 km buffer surrounding a firm's geocoded location, cropped to

[8] Fathom-Global 2.0 is based on LISFLOOD-FP, a two-dimensional hydrodynamic model designed to simulate floodplain inundation over complex topography (Bates 2010). The key datasets used are the MERIT-Hydro global hydrography dataset and the MERIT-DEM global terrain dataset, which have been corrected for urban developments (Yamazaki et al. 2017; Yamazaki et al. 2019), as well as a database of flood defense infrastructure such as dams and levees.

erase the baseline water layer. The distribution of firms' flood risk for each return period is shown in Supplemental Appendix Figure A.4. As expected, longer return periods are associated with more density in the right tail, and in each case the distribution for firms that are ever flooded during the sample is rightward shifted. Similar patterns are evident for the distribution of flood risk of routes connecting firm pairs, calculated as the average Fathom flood risk depth index of all edges along the route, weighted by edge length, as shown in Supplemental Appendix Figure A.5.

II. Floods and Supply Chain Disruption

In this section, we present motivating evidence for the disruptive impacts of flooding on firm activity and road transportation. Understanding how flooding of firms and roads disrupts firm operations is an important outcome in its own right that has received limited empirical attention in developing country contexts (Hu et al. 2019; Rentschler et al. 2021; Zhou and Botzen 2021). This is especially pertinent in Pakistan given the country's extreme vulnerability to acute flooding and underdeveloped disaster insurance market.[9] Analyzing the dynamics of the direct destructive effects of floods will also help in interpreting the results of our subsequent analysis of firm adaptation.

A. *Direct Impacts of Firm Flooding*

We first consider the impact of flooding of a firm's premises on its sales and purchases. These are "direct" impacts of flooding, in the sense that they affect the firm via its own production, in contrast to our later examination of "indirect" effects via supply chain linkages. We use the following specification:

$$(1) \qquad y_{it} = \sum_{\substack{\tau=-12 \\ \tau \neq -2}}^{12} \beta_\tau FloodExtent_{i,t-\tau} + \alpha_{im(t)} + \alpha_{iy(t)} + \alpha_t + \varepsilon_{it},$$

where y_{it} denotes log declared aggregate monthly sales or purchases for firm i in month-year t; and $FloodExtent_{i,t}$ is the maximum share of firm i's 2 km buffer that is flooded during month-year t. $\alpha_{im(t)}$, $\alpha_{iy(t)}$, and α_t are firm-month, firm-year, and month-year fixed effects, respectively, which control for firm-specific seasonality, firm-specific yearly shocks, and aggregate time trends.[10] Event dummies are binned at each endpoint of the event window (i.e., for lags and leads beyond 12 months). Standard errors are clustered at the firm level. As we observe multiple instances of flooding for a small share of firms, we restrict attention to each firm's first observed flooding event-month during the study period in this and all subsequent specifications.

[9] An estimated 3 percent of damages caused by flooding and earthquakes is covered by insurance and risk retention funds in any given year (ADB 2021).

[10] Results are robust to replacing month-year fixed effects with district-month-year fixed effects or Fathom flood risk decile-month-year fixed effects, as demonstrated in Supplemental Appendix Figures A.6 and A.7, respectively. In Supplemental Appendix F.1, we consider the robustness of results to using alternative estimators to address potential challenges associated with two-way fixed effects regressions including treatment lags and leads with variation in treatment timing. Very similar results obtain when controls for suppliers' flooding are also included.

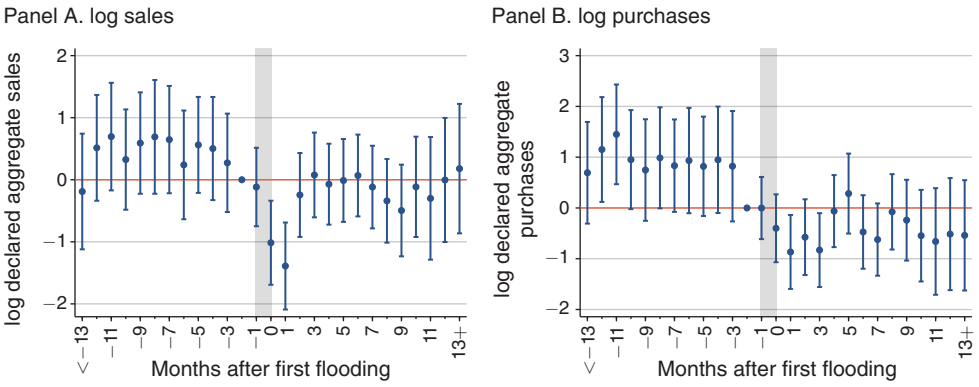


FIGURE 3. IMPACT OF FLOODING ON FIRM SALES AND PURCHASES

Notes: The panels plot OLS estimates of the effect of flooding on log declared sales and purchases as specified in equation (1). The unit of observation is a firm-month-year. The 95 percent confidence intervals rely on standard errors clustered at the firm level.

We choose the period two months before the firm's first recorded flood as the omitted reference period and shade the period from $\tau = -1$ to $\tau = 0$ as the period during which the firm is likely to have first experienced flooding. This reflects the fact that there is a lag between the onset of flooding and the date at which UNOSAT satellites capture flood extents.

The results of estimating this specification, shown in Figure 3, display reductions in both the sales and purchases of flood-hit firms in the direct aftermath of flooding.[11] The immediate impacts are statistically significant and economically large. While the reported coefficients display the estimated effect for a unit change in the maximum share of the firm's 2 km buffer flooded (i.e., for a buffer moving from not at all flooded to entirely flooded), this is a much larger flooding extent than observed in practice. To benchmark the magnitude of the coefficient estimates in this and subsequent specifications, we therefore discuss effect sizes for both the mean treated firm across firms experiencing *any* flooding (which typically experiences flooding of only a small share of its 2 km buffer) and for a severely flooded firm that experiences flooding of 10 percent of its buffer area (an acute level of flooding only experienced by 1 percent of firms).

During the month of impact, for the mean treated firm—which sees only 1.3 percent of its 2 km buffer flooded—sales decline by 1.3 percent and purchases by 0.5 percent. Impacts are much stronger for firms that experience severe flooding, with firms whose buffers are 10 percent flooded experiencing sales declines of 9.7 percent and purchases declines of 3.9 percent (a 0.05 and 0.02 standard deviation decline in log sales and log purchases, respectively). Supplemental Appendix Figures B.1 and B.2 show results where $FloodExtent_{it}$ is replaced by an indicator variable capturing whether a firm sees

[11] Reassuringly, trends are flat in the full pretreatment window for sales outcomes and in the period up to two months before recorded flooding for purchases outcomes. The slight decline in purchases in the month before flooding, while insignificant, may reflect anticipatory contractions of purchases once imminent floods are forecast, as well as lags between the onset and satellite capture of flooding.

0–5 percent, 5–10 percent, or more than 10 percent of its buffer flooded; these suggest that strongly flooded firms see their sales and purchases decline most. Supplemental Appendix Figure B.3 reveals that flooding depresses sales and purchases broadly across sectors, with the most pronounced effects for manufacturing, retail, and wholesale firms, and intuitively weaker effects for services. While it is challenging to make comparisons across settings and shock types, the magnitude of these impacts is in line with estimated effects from natural disaster and conflict shocks in other contexts (Gröschl and Sandkamp 2023; Del Prete, Di Maio, and Rahman 2023).[12]

These large immediate declines are relatively short-lived. Sales and purchases quickly revert to close to the preexposure levels: While the point estimates are slightly lower than preflood levels, the magnitudes are small, and no coefficients are statistically significantly different from preexposure levels for either outcome from four months after flooding. Given this significant recovery relative to the immediate decline, the small and statistically insignificant longer-term reductions are unlikely to be driven by continued direct disruptive impacts of the shock, but may reflect flooded firms (or their supply chain partners) undertaking adaptive adjustments, which are, on average, costly (consistent with the empirical results in Section III and the model in Section IV). Reversion to preflood levels is especially clear for manufacturing firms and firms that rely less heavily on relationship-specific inputs and which may therefore be less likely to incur the costs of adaptation.

Given these sizeable impacts of flooding on firm operations, Supplemental Appendix Figure A.9 considers whether floods are sufficiently disruptive as to result in the exit of the worst-affected firms. The results in the pooled sample show a modest but statistically insignificant positive impact on firm exit in the direct aftermath of flooding, while strong effects are observed for some individual flood events.

B. *Direct Impacts of Road Flooding*

Flooding may disrupt firm and supply chain network activity not only via direct damage to firm buildings, equipment and stocks, but also as a result of disruptions to the road network. Our GPS data allow us to study the firm- and network-level effects of such disruptions given the fine-grained lens they provide into flood-induced road closures. Given that roads closed due to flooding are often reopened rapidly, we examine flood-induced road disruptions at the weekly level, with treatment centered around the first week of observed flooding in a particular year, using the following specification:

$$(2)\ y_{iw} = \sum_{\substack{\tau=-10 \\ \tau\neq -2}}^{20} \beta_\tau \cdot \mathbf{1}\big(i \textit{ flooded at } w-\tau \text{ and } w-\tau \in y(w)\big) \cdot \textit{FloodExtent}_{i,y(w)} + \alpha_i + \alpha_{dw} + \varepsilon_{iw},$$

[12] Our firm data do not include profits, costs, quantities or prices, so we cannot estimate impacts on these alternative measures of firm operations directly. In Supplemental Appendix Figure A.8, we examine impacts on profits using available data on sales and materials purchases and parameter estimates from De Loecker, Goldberg, Khandelwal, and Pavcnik (2016). We find that profits are temporarily depressed following floods, driven by lower sales with a relatively flat sales-to-purchases ratio.

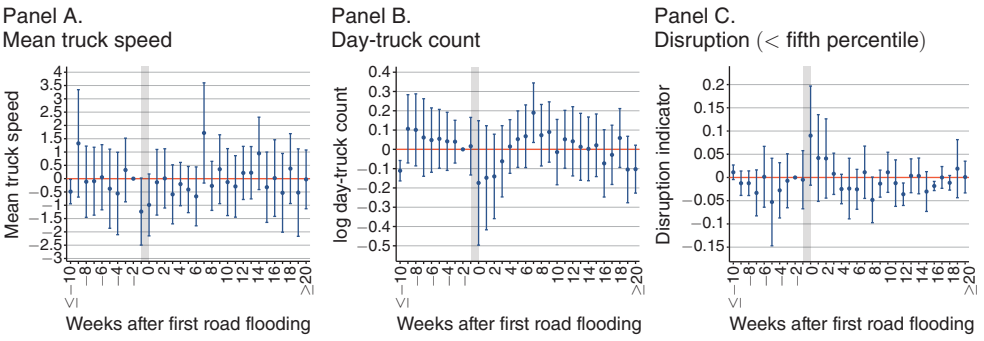


FIGURE 4. IMPACT OF FLOODING ON ROAD TRAFFIC

Notes: The panels plot OLS estimates of the effect of road flooding on different traffic outcomes following equation (2). The unit of observation is a road edge–week. Panel A excludes observations with day-truck counts < 10. Panels B and C drop edges for which the first percentile of day-truck counts is zero. The 95 percent confidence intervals rely on standard errors clustered at the district-time level. Control group means are 31.6 km/h (mean speed), 4.1 (average log number of trucks), and 5 percent (average probability of disruption).

where y_{iw} is an outcome for road edge i during week w; $y(w)$ is the year of week w; $FloodExtent_{i,y(w)}$ is the share of the total road length of i that is flooded in the first week of flooding during $y(w)$; α_i are road edge fixed effects; and α_{dw} are district-week fixed effects.

To capture alternative measures of road disruption, we consider several outcome variables. Panel A of Figure 4 shows the impact on the mean speed of trucks traveling on the edge; panel B on the log day-truck count; and panel C on an indicator variable denoting whether the road edge is "disrupted," defined as having a day-truck count on the edge that is lower than the fifth percentile for the relevant edge across all weeks.[13]

The results of these specifications paint a consistent picture of sizable but brief disruptions to traffic induced by flooding of roads. Mean truck speeds at the road edge–week level decline by a marginally statistically significant 1 km/hour in the week in which flooding is recorded and return to preflooding levels by the following week. Point estimates for day-truck counts, shown in panel B, show a decline in the range of 14–17 percent, with reversion to preflooding levels within a month of flooding. A disruption indicator based on a threshold of the fifth percentile of edge-level day-truck counts shows an increase of 9 percentage points in the week in which flooding is recorded, with reversion within a fortnight.

The results in this section suggest that floods have sizable disruptive impacts on firm operations and road traffic. While effects on firm exit are likely mostly permanent, the direct disruptive impacts of floods on intensive margin sales and purchases as well as road traffic are transient, lasting for a matter of only months or weeks, respectively. This dynamic pattern is informative for our understanding of potential

[13] The day-truck count is the number of different trucks traveling on a given edge during a given week, counting each truck more than once if they travel on the edge on different days of the week. For the specification examining mean speed in panel A, we consider the set of edge-week observations for which we have at least ten day-truck observations with valid speed since mean speed is poorly measured when trucks pass very infrequently. The regressions shown in panels B and C of Figure 4 exclude edges where the first percentile of the day-truck counts is zero; these are edges that are infrequently traversed by trucks in our dataset.

adaptive responses: If firms' changes in production choices show more persistence than the short-lived direct disruptive effects of flooding, these long-term responses could reflect firms' changed understanding of the risk they face.

III. Evidence for Firm and Supply Chain Adaptation

In this section, we turn to the key question of whether firms undertake actions following flood events that are adaptive in the sense of reducing their vulnerability to future flooding. We consider several potential margins along which firms may reduce their future flood risk in the aftermath of flood exposure. First, flooded firms may relocate toward areas that are less exposed to flood risk. Second, firms may adjust their choice of supply partners to lower indirect flood exposure, either via diversification by transacting with a larger number of supply partners, or by shifting toward less flood-prone suppliers. Finally, firms may respond to flooding of key supply routes by reducing their dependence on supply partners reached via flood-prone routes.

These adaptation margins may capture both firms' forward-looking decision-making to reduce their future vulnerability to flooding, and mechanical reductions in flood risk if, for instance, flooded firms (which have higher average flood risk) see reductions in flood risk on average if they move, even if they relocate randomly across space. Both forward-looking and mechanical responses reduce firms' flood risk and are therefore of interest in understanding firm adaptation to flood risk. Separating mechanical from forward-looking adaptation is, however, important in considering policies that may help to facilitate adaptation. In Sections IIIC and IIID, we provide evidence that isolates forward-looking adaptive behavior in supplier and route choice, and hence suggests that adaptive responses at least in part reflect forward-looking actions to reduce future flood risk.

A. *Location Choice*

We first consider the impact of flooding on firm relocation decisions for the 60 percent of firms for which we have a geocoded firm location in both 2011 and 2019. In these specifications, we consider whether flooding induces firms to relocate and how far flooding prompts firms to move toward less flood-prone locations.

Firm locations are geocoded from address strings associated with each firm in 2011 and 2019. Small differences in the address strings (for instance, the same street address being entered with and without a building number) may result in different geocodes being assigned in the two years even when a firm has not moved. Summary statistics in Supplemental Appendix Table A.3 reveal that there is a nonzero difference between the 2011 and 2019 location of 68 percent of firms, an implausibly high relocation rate over an eight-year horizon, which is likely predominantly accountable to these small discrepancies in address information. Defining firm relocation based on a threshold of 10 km gives a more plausible relocation rate of 17 percent, so we use this as the threshold for defining a firm "move."[14]

[14] Results are consistent using other thresholds, as shown in Supplemental Appendix Tables A.4 to A.6. Supplemental Appendix Table A.7 also shows the robustness of results to restricting attention to firms that only have one listed address.

Table 1—Impact of Flooding on Firm Relocation and Location Flood Risk

	Move dummy		Δ Flood risk (cm)	
	(1)	(2)	(3)	(4)
Max share of 2 km buffer flooded	2.504	3.081	−197.2	−32.23
	(0.845)	(0.912)	(113.0)	(46.43)
Average effect of mean flooded buffer	0.394 pp	0.473 pp	−3.804 cm	−0.622 cm
Average effect of 10% flooded buffer	3.668 pp	4.467 pp	−19.722 cm	−3.223 cm
District FE	Yes		Yes	
District × Fathom 1 in 100 FE		Yes		Yes
R^2	0.033	0.054	0.102	0.448
Observations	43,847	43,454	7,474	7,328

Notes: Columns 1 and 2 display logit estimates of the effect of flooding on a 10 km-relocation indicator following equation (3). Columns 3 and 4 report OLS estimates of the effect of flooding on moving firms' change in flood risk as specified in equation (4). Observations are firms fully geocoded in 2011 and 2019. The flood risk regressions only include firms that moved by $>$ 10 km. Standard errors (in parentheses) are clustered at the district level. Average effect of mean (10 percent) flooded buffer refers to the average estimated effect for a firm experiencing the mean level of flooding among treated firms in the estimation sample (flooding of 10 percent of its 2 km radius buffer). R^2 refers to McFadden's pseudo R^2 for columns 1 and 2.

The following logit specification is used to examine the impact of flooding on the probability of firm relocation during the study period:

$$Pr(Move_i) = F(\beta FloodExtent_i + \alpha_{zd}), \tag{3}$$

where $Move_i$ is an indicator denoting whether firm i moved during the sample period, and $FloodExtent_i$ is the maximum share of firm i's 2 km buffer that is flooded during the firm's first experienced flood during the study period. We consider specifications including district-level fixed effects α_d and fixed effects α_{zd} at the level of *district* × *decile of flood risk* (with a 1-in-100-year return period) in order to restrict attention to within-district variation among firms with similar levels of underlying flood risk. Standard errors are clustered at the district level.

The results, shown in columns 1 and 2 of Table 1, suggest that firm flooding increases the probability that firms relocate during the sample period. Given that no firms in the sample see flooding of the entirety of their 2 km buffer ($FloodExtent_i = 1$, which corresponds to the main estimated effect in Table 1), we again report effect sizes for both the mean treated firm among those experiencing any flooding, and a severely flooded firm experiencing inundation of 10 percent of its buffer. In the specification with district fixed effects (*district* × *Fathom flood risk decile* fixed effects), mean flooding, which inundates 1.15 percent of a firm's buffer, increases the relocation probability by 0.39 percentage points (0.47 percentage points) on average, relative to the 17.34 percent probability of moving by more than 10 km among nonflooded firms. For firms that experience severe flooding affecting 10 percent of their buffer, there is a much stronger effect of 3.67 percentage points.

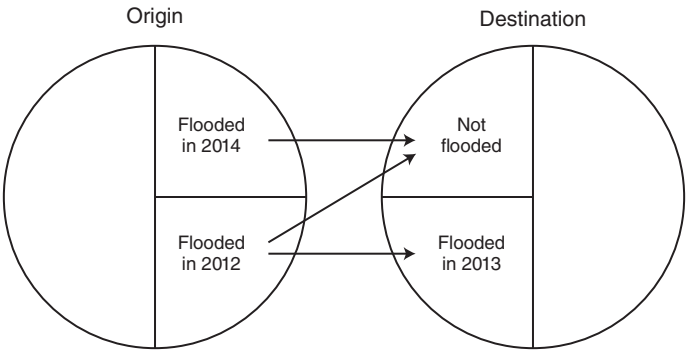


FIGURE 5. ILLUSTRATION OF DIFFERENTIAL RELOCATION BASED ON DESTINATION FLOOD HISTORY

Firm relocation may help firms to adapt to flood risk if they relocate toward areas with lower underlying flood risk. We investigate this using the following specification:

$$\Delta FloodRisk_i = \beta FloodExtent_i + \alpha_{zd} + \varepsilon_i, \tag{4}$$

where $\Delta FloodRisk_i$ measures the change in Fathom flood risk between firm i's 2019 and 2011 addresses in units of expected flood depth under a 1-in-100-year flood; with $FloodExtent_i$ and α_{zd} as above. Standard errors are clustered at the district level.

The results, shown in columns 3 and 4 of Table 1, suggest that flooding indeed induces firms to relocate to less flood-prone locations. Using within-district variation, the mean treated firm that moved more than 10 km (which has 1.92 percent of its buffer flooded) sees a 3.8 cm reduction in expected 1-in-100-year flood depth, while severely flooded firms see a reduction of 19.7 cm. These changes correspond to large percentage reductions relative to the average expected 1-in-100-year flood depth of 37 cm among firms that experience any flooding during the sample period. Column 4 of Table 1 also displays a negative effect when restricting to within-district risk decile variation, with an intuitive reduction in magnitude and statistical significance. Supplemental Appendix Table B.1 shows that smaller firms, which intuitively may be expected to be more footloose, are differentially likely to relocate toward less flood-prone areas following floods.[15]

In a final specification relating to relocation, we consider evidence that relocating firms take into account recent flood history in deciding on a *destination* location. Intuitively, this specification tests whether flooded firms that relocate during the sample period are more likely to move to destination areas that are flooded if, at the time when the relocating firm's area was flooded, the destination area had not yet been flooded.[16] This is illustrated in Figure 5: Restricting attention to firms that

[15] This pattern of heterogeneity is not driven by a higher propensity for adaptive relocation among young firms, single versus multibranch firms, or firms in industries that on average have lower fixed assets.

[16] Recall that firm addresses are only observed at the beginning and end of the study period, so we do not observe when a relocating firm moves. The strategy of this regression assumes that firms are more likely to move after having been flooded, consistent with the evidence in Table 1. If that was not the case, the coefficient in equation (5) is unlikely to be nonzero. For locations in destination districts that are never flooded, t_d is set to a time period after the sample period. We drop origin locations not flooded during the study period, and location pairs without relocation flows.

Table 2—Impact of Destination Flood History on Relocation Flows

	Number of firms moved
Dest. flooded 12 months prior	−1.076
	(0.248)
Origin × Destination FE	Yes
Origin × Flood event (month) FE	Yes
Flood event of destination FE	Yes
Move distance restriction	> 10 km
Observations	1,593

Notes: The table reports Poisson pseudo-maximum-likelihood estimates of the effect of destination flood history on relocation flows following equation (5). Standard errors (given in parentheses) are clustered at the origin-destination level. The sample is restricted to firms whose 2011 and 2019 locations are known and > 10 km apart.

relocate from the same origin district to the same destination district, do we see that firms flooded in 2014 (who were in a position to have witnessed 2013 flooding) are less likely than those flooded in 2012 (who had not witnessed 2013 flooding) to relocate to areas of the destination district flooded in 2013?

We examine this using the following gravity Poisson specification:

$$X_{ot_o dt_d} = \alpha_{od} + \alpha_{ot_o} + \alpha_{dt_d} + \beta \mathbf{1}(t_o - t_d > 12) + \varepsilon_{ot_o dt_d}, \tag{5}$$

where the unit of observation $ot_o dt_d$ is the area of an origin district o first flooded in year-month t_o paired with the area of a destination district d, which was either first flooded in year-month t_d or never flooded. $X_{ot_o dt_d}$ denotes relocation flows capturing the number of firms moving by more than 10 km from area ot_o to area dt_d; α_{od} are *origin district* × *destination district* fixed effects; α_{ot_o} are fixed effects for the area of origin district o flooded at time t_o; α_{dt_d} are fixed effects capturing destination areas flooded at time t_d; and $\mathbf{1}(t_o - t_d > 12)$ is an indicator that takes the value one if the flooding of area ot_o postdates that of area dt_d by more than 12 months. Standard errors are clustered at the level of origin-destination district pairs.

The results of this analysis in Table 2 suggest that, within origin-destination district pairs, firms relocating from origin district regions that are flooded more than 12 months after destination district regions are indeed one-third as likely to relocate to the latter regions as earlier-flooded firms. Firms therefore appear to take past flooding of destination locations into account when deciding where to move, systematically avoiding those destination regions that they have seen flooded.[17]

[17] Supplemental Appendix Table A.8 reports results for equivalent specifications where we consider firms relocating from origin district regions that are flooded more than 24 or 36 months (rather than 12 months) after destination district regions. The finding that firms take past flooding into account when deciding where to move remains robust in these specifications. This suggests that flooding has very persistent impacts on in-migration of firms and reduces the likelihood that results are driven by, for instance, reduced firm profitability or labor market effects in the aftermath of flooding. Consistent with this, analysis of firm-entry patterns at the district level reveals depressed entry in the month following flooding, but quick recovery thereafter.

B. *Diversification of Suppliers*

In addition to location choice, another margin along which firms might adapt to flood risk is in their choice of supply partners. If flooding increases the risk that a firm's suppliers will be unable to meet their commitments, firms may hedge this risk by diversifying their supplier base or shifting toward less flood-prone suppliers.[18] We first consider whether firms adapt to flood risk by increasing the number of suppliers from which they source. This may help to reduce dependence on individual suppliers and spread risk across suppliers with uncorrelated shocks, in line with a literature examining diversification in other contexts (Cole, Elliott, Okubo, and Strobl 2019; Meltzer, Ellen, and Li 2021; Boehm and Sonntag 2023; Castro-Vincenzi 2024; Castro-Vincenzi, Khanna, Morales, and Pandalai-Nayar 2024).

We test whether firms diversify their supplier base in response to flooding of their own premises using the following specification:

$$y_{it} = \sum_{\substack{\tau=-12 \\ \tau\neq-2}}^{12} \beta_\tau \mathit{FloodExtent}_{i,t-\tau} + \alpha_{im(t)} + \alpha_{iy(t)} + \alpha_t + \varepsilon_{it}, \tag{6}$$

where y_{it} denotes firm i's log number of suppliers in month-year t; $\mathit{FloodExtent}_{i,t}$ is the maximum share of firm i's 2 km buffer that is flooded during month-year t; and $\alpha_{im(t)}$, $\alpha_{iy(t)}$, and α_t are firm-month, firm-year, and month-year fixed effects respectively. Standard errors are clustered at the firm level.[19]

It may be more intuitive to expect firms to diversify suppliers in response to flooding of the suppliers themselves rather than their own premises. We test this using the following specification, where the coefficients of interest are the $\beta_{1,\tau}$ terms, including controls for the firm's own flood status:

$$y_{bt} = \sum_{\substack{\tau=-12 \\ \tau\neq-2}}^{12} \beta_{1,\tau} \mathit{SellerFlood}_{b,t-\tau} + \sum_{\substack{\tau=-12 \\ \tau\neq-2}}^{12} \beta_{2,\tau} \mathit{OwnFlood}_{b,t-\tau} + \alpha_{bm(t)} + \alpha_{by(t)} + \alpha_t + \varepsilon_{bt}, \tag{7}$$

where y_{bt} denotes the log number of suppliers of buyer firm b during month-year t. $\mathit{SellerFlood}_{b,t}$ are the treatment terms, based on the firm's first observed supplier flooding event. Given that many firm-pairs transact only infrequently (see Section IA), a buyer may be affected by flooding of those suppliers from which it sources but with which it happens not to transact in the month under consideration. In constructing the treatment variable, we therefore define a buyer firm's suppliers as those firms from which the buyer firm has made any purchases in the prior three

[18] All specifications investigating supplier choice restrict attention to firms that did not relocate more than 10 km over the sample period, in order to remove the potential effects of relocating firms switching suppliers to those based in their new location. All results are robust to including relocating firms, as shown in Supplemental Appendix F.9. Supplier diversification results further restrict the sample to cases where buyer and seller reports coincide exactly.

[19] The findings are robust to instead considering as an outcome variable an alternative measure of diversification given by the inverse Herfindahl index, defined by $\left(\sum_{i=1}^{N} \left(\mathit{Share\ of\ Purchases}_i\right)^2\right)^{-1}$, where N is the total number of suppliers, as shown in Supplemental Appendix Figure A.10.

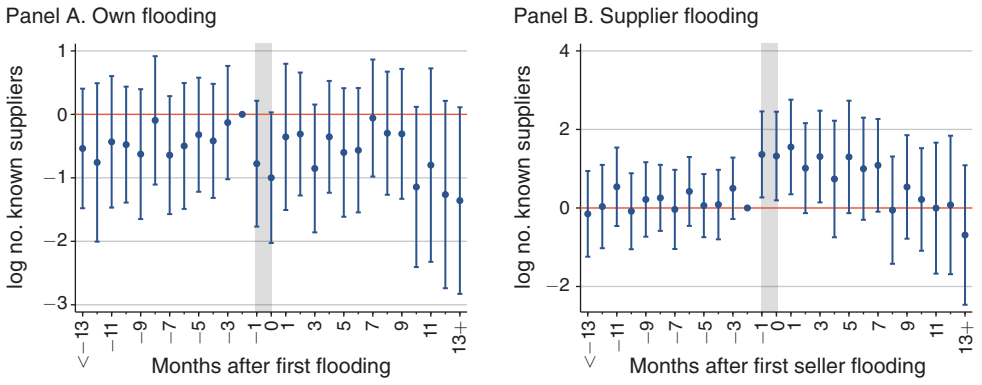


FIGURE 6. SUPPLIER DIVERSIFICATION: IMPACT OF FLOODING ON LOG NUMBER OF SUPPLIERS

Notes: Panels A and B plot OLS estimates of the effect of own flooding and supplier flooding on the log number of suppliers following equations (6) and (7), respectively. Observations are firm-month-years whose 2011 and 2019 addresses are known and $\leq$ 10 km apart. We restrict attention to transactions for which buyer and seller reports coincide precisely. The 95 percent confidence intervals rely on standard errors clustered at the firm level.

months.[20] $SellerFlood_{b,t-\tau}$ is the maximum share of the 2 km buffer flooded across all suppliers that account for more than 10 percent of firm b's purchases within the three-month window. $OwnFlood_{b,t}$ are controls for the firm's own flood status during the first observed supplier flooding event, based on the maximum share of firm b's 2 km buffer that is flooded during month-year t. $\alpha_{bm(t)}, \alpha_{by(t)}, \alpha_t$ are as previously, and standard errors are clustered at the firm level.

The results of both specifications are shown in Figure 6. The specification from equation (6) finds no evidence for diversification in response to flooding of a firm's own premises (panel A). Firms do, however, temporarily increase the number of their suppliers in response to supplier flooding (panel B). For the mean treated firm, whose maximally flooded supplier sees 1.36 percent of its buffer flooded, the number of suppliers increases by 1.81 percent by three months after supplier flooding. Severely affected firms, with a maximum flooded-buffer share of 10 percent among their suppliers, see a larger increase of 14.1 percent. The response is relatively short-lived, with reversion to preexposure levels within a year. This is consistent with firms making up for missing or lost inputs by temporarily purchasing from new suppliers, but does not indicate that firms make long-term adaptive adjustments by diversifying their supplier base.

Beyond changes in the *number* of a firm's suppliers, another potential margin of adaptation is to change the *characteristics* of their suppliers by shifting toward a portfolio of suppliers less prone to flooding. We consider two dimensions of such a decision. In the next subsection, we examine how far flood-affected firms shift the composition of their suppliers toward suppliers located in less flood-prone regions. In the subsequent subsection, we consider whether floods affecting transportation routes also induce firms to reduce dependence on suppliers reached via flood-prone routes.

[20] All results that use this assumption are robust to alternatively defining a buyer firm's suppliers based on a 6- or 12-month window; see Supplemental Appendix F.8.

C. *Flood Risk of Suppliers*

Does exposure to flooding induce firms to source from less flood-prone suppliers? We start by demonstrating that there is significant potential for firms to adapt by choosing less flood-prone suppliers, and then demonstrate that this is an important margin used for adaptation in practice.

We consider the potential for firms to adapt via their choice of suppliers by examining how likely it is that firms whose suppliers are affected by flooding can easily substitute toward safer suppliers. Supplemental Appendix Table A.11 reports the probabilities that alternative *nonflooded* and *less flood-prone* suppliers in the same two-digit industry exist within 25, 50, and 100 km of the buyer firm. We also consider separately this probability restricting attention to suppliers in the same two-digit industry that are both less flood prone and larger, to avoid potential concerns that, for instance, smaller firms could not produce inputs of correspondingly high quality. The results reveal substantial potential for adaptation: For instance, there is a 90 percent probability that a nonflooded alternative supplier in the same industry exists within 25 km, and a 92 percent (77 percent) probability that a less flood-prone (less flood-prone and larger) alternative supplier in the same industry exists within 25 km.

In order to test how far flood-affected firms exploit this potential by shifting their sourcing toward less flood-prone suppliers, we study changes in the risk profile of a buyer firm's suppliers around the time of flooding of the buyer itself, or flooding of any of their suppliers (again based on the preceding three-month window). This reflects the fact that firms deciding on the risk profile of their supplier base may take into account both their experience of supply chain disruptions caused by flooded suppliers, and their own direct experience of floods. We compare changes in the supplier mix of flood-exposed firms relative to those that are not flood exposed, around the time of a flood, using the following empirical specification:

$$\Delta y_{bt^*} = \beta_1 \mathit{OwnFlood}_{bt^*} + \beta_2 \mathit{SellerFlood}_{bt^*} + \alpha_{d(b)t^*} + \epsilon_{bt^*}, \tag{8}$$

where t^* denotes the month-year of a flood event; $\mathit{OwnFlood}_{bt^*}$ is the maximum share of buyer b's 2 km buffer that is flooded at t^*; $\mathit{SellerFlood}_{bt^*}$ is the maximum share of the 2 km buffer flooded at t^* across all sellers, which account for at least 10 percent of b's purchases over the previous three months; and $\alpha_{d(b)t^*}$ are *buyer district* $\times$ *time* fixed effects. We additionally report results using *buyer district* $\times$ *time* $\times$ *buyer 1-in-100-year flood risk decile* or *buyer district* $\times$ *time* $\times$ *buyer industry* fixed effects.

The dependent variable, Δy_{bt^*}, captures the change in the sales-weighted average flood risk of all of b's suppliers in the three months before versus after flood exposure, given by

$$\Delta y_{bt^*} = \frac{\sum_{s \in S_b(t^*, t^*+3]} \sum_{t \in (t^*, t^*+3]} \mathit{Sales}_{bst} \mathit{Risk}_s}{\sum_{s \in S_b(t^*, t^*+3]} \sum_{t \in (t^*, t^*+3]} \mathit{Sales}_{bst}} - \frac{\sum_{s \in S_b(t^*-3, t^*]} \sum_{t \in (t^*-3, t^*]} \mathit{Sales}_{bst} \mathit{Risk}_s}{\sum_{s \in S_b(t^*-3, t^*]} \sum_{t \in (t^*-3, t^*]} \mathit{Sales}_{bst}}, \tag{9}$$

Table 3—Impact of Flooding on Supplier Flood Risk

	Δ Supplier flood risk (cm)		
	(1)	(2)	(3)
Own max flood extent	−8.139	−7.542	−12.18
	(9.827)	(7.001)	(12.25)
Suppliers' max flood extent	−60.41	−61.63	−72.52
	(16.13)	(16.66)	(18.60)
Average effect of mean flooded supplier buffer in cm	−0.895	−0.913	−1.075
Average effect of 10% flooded supplier buffer in cm	−6.041	−6.163	−7.252
Time × District FE	Yes		
Time × District × Risk decile FE		Yes	
Time × District × Industry FE			Yes
R^2	0.0117	0.0303	0.0623
Observations	137,957	137,192	132,854

Notes: The table reports OLS estimates following equation (8) of the effects of supplier and own flooding on the change in sales-weighted average flood risk among all suppliers. Observations are firm-by-flood-year-month pairs whose 2011 and 2019 addresses are known and ≤ 10 km apart. Standard errors (in parentheses) are clustered at the *time* × *district* level. Effect of mean (10 percent) flooded supplier buffer refers to the estimated effect of a firm experiencing average, across treated firms in the estimation sample, (10 percent) supplier flooding. Risk decile indicates the decile of a firm's own flood risk for a 1-in-100-year return period.

where $Risk_s$ indicates seller s's expected flood depth under a 1-in-100-year flood; $Sales_{bst}$ represents the total sales from seller s to buyer b in year-month t; and $S_b(t_1, t_2]$ is the set of b's suppliers over $(t_1, t_2]$.

The results are shown in Table 3. Buyers do not appear to adjust the flood risk composition of their suppliers in response to flooding of their own premises, but do respond to flooding of their suppliers by shifting toward less flood-prone suppliers. The magnitudes are sizable and quite consistent across specifications including different fixed effects. In the central specification including district × time fixed effects, the mean treated observation (which sees a maximum flood extent among its sellers' buffers of 1.48 percent) experiences a 0.9 cm reduction in the sales-weighted average supplier flood risk for a 1-in-100-year flood event. Where suppliers are more severely flooded (with a supplier maximum flooded buffer share of 10 percent), the sales-weighted average supplier flood risk decreases by 6.04 cm. This compares to a sales-weighted average flood risk among treated firms' suppliers of 24.97 cm.

The shift toward less flood-prone suppliers may be driven by forward-looking adaptation, but is also consistent with the mechanical effect of being forced to source less from flood-hit suppliers while their operations are disrupted. To disentangle these effects, we consider whether flooding induces firms to shift toward less flood-prone suppliers among the subset of their suppliers that are not flooded, and therefore for which such mechanical effects are shut down. This provides the first opportunity to isolate adaptation that derives from firms' forward-looking decision-making to reduce future vulnerability to flooding.

Table 4 estimates equation (8) where the dependent variable is calculated as the change in sales-weighted average flood risk of buyer b's suppliers that are not hit

Table 4—Impact of Flooding on Flood Risk of Nonflooded Suppliers

	Δ Supplier flood risk (cm)		
	(1)	(2)	(3)
Own max flood extent	−13.89	−9.281	−17.83
	(9.852)	(6.398)	(12.26)
Suppliers' max flood extent	−25.40	−25.71	−24.19
	(9.463)	(9.202)	(10.22)
Average effect of mean flooded supplier buffer in cm	−0.376	−0.381	−0.358
Average effect of 10% flooded supplier buffer in cm	−2.540	−2.571	−2.419
Time × District FE	Yes		
Time × District × Risk decile FE		Yes	
Time × District × Industry FE			Yes
R^2	0.0094	0.0293	0.0610
Observations	137,822	137,057	132,724

Notes: The table reports OLS estimates following equation (8) of the effects of supplier and own flooding on the change in sales-weighted average flood risk among suppliers that are not flooded (defined as $\leq$ 5 percent overlap of the 2 km buffer and the flood polygon). Observations are firm-by-flood-year-month pairs whose 2011 and 2019 addresses are known and $\leq$ 10 km apart. Standard errors (in parentheses) are clustered at the *time* × *district* level. Effect of mean (10 percent) flooded supplier buffer refers to the estimated effect of a firm experiencing average, across treated firms in the estimation sample, (10 percent) supplier flooding. Risk decile indicates the decile of a firm's own flood risk for a 1-in-100-year return period.

by a flood shock.[21] While the coefficient on the supplier treatment is smaller than in Table 3 (meaning that some of the reduction in supplier risk is driven by reduced purchases from the flooded supplier), it remains large and statistically significant. This suggests that, when a buyer experiences flooding of any of their suppliers, this induces them to shift toward safer suppliers even among the subset of their suppliers not disrupted by flooding. In terms of magnitudes, the mean treated firm sees a 0.38 cm reduction in the sales-weighted average flood risk among its nonflooded suppliers, while this reduction is 2.54 cm for a more severely affected firm whose suppliers' maximum flooded buffer share is 10 percent.

Heterogeneity in firms' propensity to shift their sourcing toward less flood-prone suppliers is examined in Supplemental Appendix B, which reveals intuitive patterns. Adaptive shifts in supplier choice are more pronounced among smaller firms (Supplemental Appendix Tables B.2 and B.3) and single-branch relative to multibranch firms (Supplemental Appendix Tables B.5 and B.6). This is consistent with smaller firms having smaller and less complex supply chains—such that changes in supplier networks may be less costly and disruptive—and with multibranch firms having greater resilience to disruption from floods via hedging risk across diversified locations and reallocating activity to unaffected plants (Castro-Vincenzi, Khanna, Morales, and Pandalai-Nayar 2024). Adaptive supplier choice is also more pronounced among firms in industries that require more relationship-specific inputs, as shown in Supplemental Appendix Tables B.8 and B.9. Here, input relationship specificity is constructed as the weighted average of relationship specificity of input

[21] Here we define suppliers that are not hit by a flood shock as those that see less than or equal to 5 percent overlap of their 2 km buffer with the flood polygons, since these firms do not experience direct disruptions to sales or purchases from flooding (see Supplemental Appendix Figures B.1 and B.2). Supplemental Appendix Table A.9 shows that qualitatively similar results are obtained using alternative thresholds of 10 percent or 1 percent.

industries as in Boehm, Oberfield, South, and Waseem (2024) (where product-level relationship specificity is from Rauch 1999); this pattern of heterogeneity is robust to instead considering the output specificity of a firm's maximally flooded supplier in order to restrict attention to cases where the disrupted input is specific. This result is intuitive if it is more difficult for firms that purchase relationship-specific goods to find suitable suppliers, so that flooding of upstream suppliers will likely be more costly.[22] Supplemental Appendix Tables B.10 and B.11 demonstrate that firms whose suppliers are affected by flooding increase their share of purchases from new suppliers, while Supplemental Appendix Tables B.12 and B.13 suggest that they also source more from suppliers with multiple branches, which may be less exposed to flooding impacts given their ability to substitute to other sites.[23]

We consider the persistence of the shift toward less flood-prone suppliers among flood-affected firms, using the following specification:

$$\Delta y_{bt} = \sum_{\substack{\tau=-12 \\ \text{s.t.}\frac{\tau}{3}\in\mathbb{Z},\tau\neq 0}}^{48} \beta_{1,\tau} SellerFlood_{b,t-\tau} + \sum_{\substack{\tau=-12 \\ \text{s.t.}\frac{\tau}{3}\in\mathbb{Z},\tau\neq 0}}^{48} \beta_{2,\tau} OwnFlood_{b,t-\tau} + \alpha_{d(b)t} + \varepsilon_{bt}, \tag{10}$$

where Δy_{bt} represents the change in sales-weighted average flood risk among suppliers of firm b (defined as in equation (9)) from the previous three-month window to the window ending in the time t of the observation (b,t) (i.e., from $(t-6, t-3]$ to $(t-3, t]$); $SellerFlood_{b,t-\tau}$ indicates the maximum share of the 2 km buffer flooded across b's suppliers (again defined as those accounting for more than 10 percent of b's purchases over the three-month window preceding the flood event) during the relevant flood event; and $OwnFlood_{b,t-\tau}$ is the share of b's buffer flooded during that event. When analyzing the effects of firms' own flooding, the relevant flood ($\tau = 0$) is the firm's first direct flood event, controlling for potential concurrent flooding of the firm's suppliers. When we instead consider the impacts of flooding of a firm's suppliers, the relevant flood ($\tau = 0$) is instead the firm's first supplier flood event, controlling for potential concurrent flood exposure at the firm's premises. $\alpha_{d(b)t}$ are buyer-district-by-time fixed effects.[24]

Given this specification, a short-lived shift toward less flood-prone suppliers would yield an initial negative coefficient of interest $\beta_{1,\tau}$ (or $\beta_{2,\tau}$), followed by positive coefficients in later time periods as the buyer reverts back toward more flood-prone suppliers. Conversely, a persistent shift would be consistent with an initial negative coefficient, without evidence of positive coefficients thereafter.

Figure 7, panel B, plots the coefficients of interest $\beta_{1,\tau}$ for rolling three-month-on-three-month windows out to four years after the date of first

[22] Dimensions of firm heterogeneity along which we do not find evidence for heterogeneous adaptive supplier choice include firm age, frequency of purchases, baseline supplier diversification, industry-level average inventory ratios, shipment frequency, and output market concentration.

[23] Firms whose suppliers are flooded do not switch toward suppliers that are more diversified, larger, or more distant from the buyer firm; nor do these firms increase their share of purchases from suppliers used by their competitors.

[24] We include ever-treated observations only at the lags of interest $\tau_{(t)} = -12, -9, \ldots, 45, 48$. Including them in every time period but omitting the treatment outside the lags of interest would confound $\widehat{\alpha_{d(b)t}}$ as these observations would effectively serve as control observations.

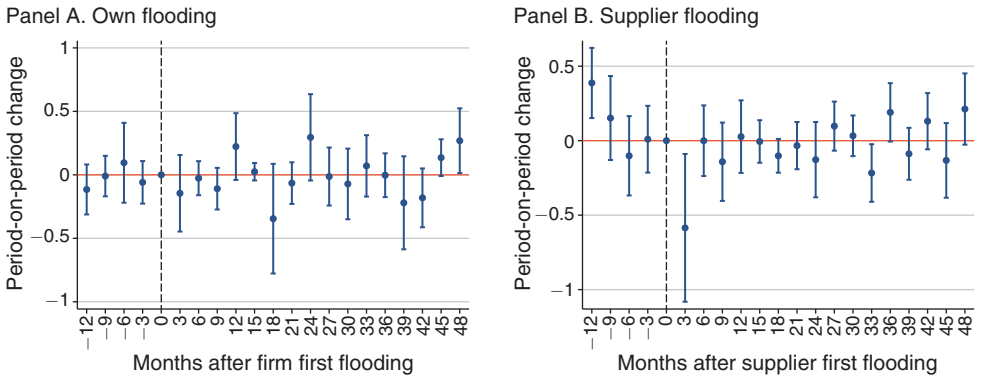


FIGURE 7. DYNAMIC IMPACT OF FLOODING ON SUPPLIER FLOOD RISK

Notes: The panels plot OLS estimates of the effect of own and supplier flooding (separately) on the change in suppliers' sales-weighted average flood risk in equation (10), following the first firm direct flooding event for panel A, and the first supplier flood event for panel B. Observations are all firm-year-months for which the 2011 and 2019 addresses are known and $\leq$ 10 km apart. The 95 percent confidence intervals rely on standard errors clustered at the *time* $\times$ *district* level.

observed supplier flooding. The results are consistent with buyer firms persistently shifting toward less flood-prone suppliers over this horizon.[25] A persistent response is also evident when restricting attention to nonflooded suppliers, as shown in Supplemental Appendix Figure A.11. Figure 7, panel A, shows no effect of flooding of a firm's own premises, consistent with the results in Table 3.

D. *Flooding of Supply Routes*

Floods may affect firm activities via transportation disruption as well as flooding of firm premises, as shown in Section IIB. We next consider whether firms adapt to flooding of transportation routes by reducing their dependence on supply partners reached via flood-prone routes.

The route-level specifications leverage the bilateral nature of transaction-level data to fully isolate adaptive behavior from potentially confounding shocks that may affect flooded firms. In particular, using variation from pairwise route-level flooding allows us to include buyer-time and seller-time fixed effects, thereby absorbing any shocks to buyers and sellers, including those that may persist even after flooded firms' sales and purchases have recovered.[26] We estimate event study regressions of the form

$$y_{bst} = \sum_{\substack{\tau=-12 \\ \tau\neq-2}}^{36} \beta_\tau \mathit{ShareRouteFlooded}_{bs,t-\tau} + \eta_{age(b,s),t} + \alpha_{bs} + \alpha_{bt} + \alpha_{st} + \varepsilon_{bst}, \tag{11}$$

[25] Supplemental Appendix Figure F.56 presents results excluding firms that experience flooding of a supplier in more than one flood event. The robustness of the results to this restriction indicates that the persistence of the effect is not driven by repeated exposures.

[26] For instance, it is possible that, even once flooded firms' sales and purchases have recovered, local labor market or correlated cost shocks (e.g., through credit markets—see Choudhary and Jain 2022) might continue to affect their operations and contribute to persistent changes in outcomes at the firm level.

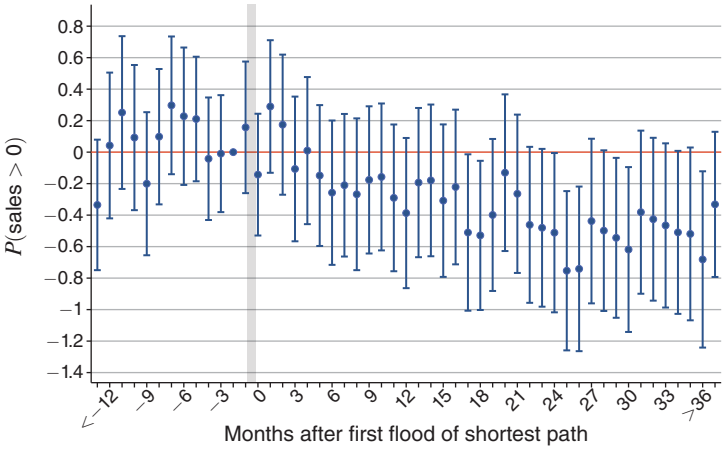


FIGURE 8. IMPACT OF ROAD FLOODING ON EXTENSIVE MARGIN SALES IN BUYER-SELLER RELATIONSHIP

Notes: The graph shows the response of the probability of sales being positive in the (b,s) relationship around the first time the shortest path between b and s gets flooded (after entry of b and s) following equation (11). The sample consists of all buyer-seller-months such that b and s are manufacturing firms, do not relocate, transact at least twice, and excludes observations (b,s,t) that are prior to the entry of b or s. The set of regressors includes $b \times s$, $s \times t$, and $b \times t$ fixed effects and months-since-first-sale dummies. The 95 percent confidence intervals rely on standard errors clustered at the relationship level.

where y_{bst} is an outcome at the buyer-seller-time level (sales in the (b,s) relationship during month-year t, or an indicator variable denoting whether sales are positive); and $ShareRouteFlooded_{bst}$ is the share of the ordinary-time (i.e., during nonflooded weeks) shortest-time route between b and s flooded at time t. We consider all events where the shortest-time route between b and s is flooded for the first time after entry of b and s. $ShareRouteFlooded_{bst}$ is calculated at the weekly level and the maximum for weeks during a given month is used to generate monthly level variables. A set of indicator variables for the age of the buyer-seller relationship, $\eta_{\text{age}(b,s),t}$, is included given evidence for strong life-cycle effects in buyer-seller relationships (see Supplemental Appendix Figure A.12). α_{bs}, α_{bt}, and α_{st} are, respectively, buyer-seller, buyer-time, and seller-time fixed effects. In the extensive margin specifications, the set of observations consists of all triples (b,s,t) where b and s transact at least twice, and b and s have both entered by time t. In the intensive margin specifications, the set of observations (b,s,t) is all triples where s has positive sales to b at t. In the baseline specification, we restrict attention to manufacturing firms to ensure that transactions correspond to shipments of physical goods; results are, however, robust to lifting this restriction (Supplemental Appendix Table F.33).

Figure 8 shows the extensive margin results. After a flood hits the route between a buyer-seller pair, the likelihood of the relationship remaining active declines compared to nonflooded relationships. Trends are flat in the yearlong window prior to flood exposure, increasing confidence that this effect is driven by the impact of flooding. To illustrate the magnitude of the estimates, a point estimate of -0.26 six months after treatment implies that the transaction probability in that period declines by 0.1 percentage points for the median flooded route (which sees 0.4 percent of its

length flooded).[27] The figure shows that the decline is persistent for at least three years, far beyond the duration of road disruptions, which typically last less than one month (see Section IIB).

Conditional on a transaction occurring, we do not find any adjustment in the transaction magnitude following a flood (Supplemental Appendix Figure A.13). This suggests that substitution away from supply partners reached by flooded routes is driven by transactions ceasing rather than intensive margin reductions in transaction volumes.

These results provide strong evidence of adaptation using clean exogenous assignment of the treatment: Controlling for any direct effects of floods on buyers or sellers themselves, as well as buyer-seller fixed effects, short-lived flood disruption of transportation routes between buyer-seller pairs results in persistent cessation of transactions between them.

E. *Robustness*

Supplemental Appendix F considers the robustness of the reduced-form evidence on adaptive behaviors to using alternative estimators that aim to overcome potential challenges with the use of two-way fixed effects regressions including treatment leads and lags (Supplemental Appendix F.1); excluding industries for which supply disruptions of the nature considered in the analysis may not be pertinent (Supplemental Appendix F.2–F.4); considering only transaction observations where buyer and seller reports coincide exactly (Supplemental Appendix F.5); using buffers of radius 1 km or 3 km, rather than 2 km, to define flood exposure and risk (Supplemental Appendix F.6); and considering floods with return periods of 1-in-10 or 1-in-50 rather than 1-in-100 years (Supplemental Appendix F.7). The central results are all qualitatively robust to these alternative specifications.

F. *Discussion of Potential Mechanisms*

The results in this section suggest that firms adapt to flood risk in the aftermath of flood events via relocation toward less flood-prone locations, diversification, and shifts in the supplier mix away from those in more flood-prone locations and reached via flooded routes. Several alternative mechanisms may underlie these adaptive responses.

One possible mechanism is that floods lower the fixed cost of making changes that the firm may already have wished to make, such as relocating to safer areas, or force firms to experiment with new suppliers, resulting in lasting changes as in Larcom, Rauch, and Willems (2017). Such a mechanism cannot, however, account for the finding in Table 4 that supplier flooding induces buyers to shift their *nonflooded* supplier mix toward less flood-prone firms, or flood-induced supplier diversification in Section IIIB.

Rational learning would suggest that firms affected by floods change their beliefs over the underlying distribution of flood risk and that adaptive actions reflect a rational

[27] For comparison, the unconditional probability of a relationship being active in our panel—meaning not before the first transaction of each firm pair—is 18.45 percent.

response to this. Such channels have been studied in a recent literature examining individual decision-making in relation to climate risks (Lybbert, Barrett, McPeak, and Luseno 2007; Moore 2017; Kala 2017; Patel 2024), though evidence from firm behavior remains scarce (Kremer, Rao, and Schilbach 2019).

Alternatively, floods may increase the salience of climate risk. Such "availability bias" might induce flood-hit firms to infer erroneously that they are subject to higher flood risk relative to a firm with identical statistical information, simply by virtue of recent experience (Tversky and Kahneman 1973; Kahneman 2011; Bordalo et al. 2023), behaviors that have been documented in individuals' decisions to purchase weather insurance and responses to surveys about climate change (Gallagher 2014; Turner, Said, and Afzal 2014; Karlan, Osei, Osei-Akoto, and Udry 2014; Deryugina 2013).

While mechanisms predicated on experience-based updating should imply persistent responses, availability heuristics would predict larger impacts from more recent floods. The evidence in Section IIIA for adaptive relocation is consistent with a permanent response. Flood-induced adaptive shifts in supplier mix persist for at least four years (Figure 7), and shifts away from suppliers reached via flooded routes for at least three years (Figure 8), without evidence of attenuation in either case. While behavior may revert over longer time scales, the fact that these adaptive responses persist for as long as is measurable in our data appears inconsistent with salience effects being first order in the medium run.

IV. Quantifying the Aggregate Impacts of Adaptation

While the reduced form results in Section III identify evidence for adaptation, a simple aggregation of these estimates is unlikely to provide an accurate assessment of their economy-wide impacts. This is because aggregate effects will reflect general equilibrium forces (for instance, firm cost increases driven by flooding of their suppliers as well as direct damages) and firm connections via multistep linkages (for instance, flood exposure of the suppliers of a firm's suppliers, and so on). In this section, we develop a quantitative spatial model of production network formation and adaptation, which models general equilibrium and indirect effects explicitly, in order to estimate the aggregate impacts of adaptation for the resilience of the economy to future floods.

A. *Theoretical Model*

The model builds on recent two-stage models of production network formation under uncertainty (Kopytov, Mishra, Nimark, and Taschereau-Dumouchel 2024) but allows for imperfectly informed firms that learn about underlying flood risk from flood shock realizations. Firms are subject to idiosyncratic and aggregate flood risk, about which they update their beliefs in response to floods. Given the challenges in distinguishing rational from heuristic updating of flood beliefs empirically (see Section IIIF), we use a flexible framework that does not require us to take a stance on the nature of firm beliefs or the learning process.

Before flood shocks are realized, firms search for suppliers, taking into account their beliefs over the flood risk of potential suppliers. Once floods have occurred,

firms then choose suppliers to minimize costs conditional on the outcomes of their search. Extreme value distribution assumptions (following Oberfield 2018 and Boehm and Oberfield 2020, 2022) yield tractable gravity expressions for sourcing shares. We use these gravity equations alongside expressions for cost indices in changes (the "exact-hat" approach pioneered by Dekle, Eaton, and Kortum 2007) to estimate how far postflood adaptation observed in our sample affects the damages imposed by subsequent floods. Full derivations are included in Supplemental Appendix D.

Model Setup.—The economy consists of N locations indexed by n. Location n contains an exogenous number of firms J_n. Each firm sells a good that is considered differentiated by the representative household, but which is perfectly substitutable with goods produced by other firms when used as an intermediate input in production.

Households.—Households have constant relative risk aversion preferences over consumption of a bundle of goods comprising individual varieties produced by firms in different locations. Households' expenditure shares on goods from different locations are given by β_n and assumed constant given our focus on firms' adaptive decisions:

$$u(q) = \frac{1}{1-\rho} q^{1-\rho},\ \rho > 0,\ q = \prod_{n=1}^{N} \left(\frac{q_n}{\beta_n}\right)^{\beta_n},\ q_n = \left(\int_{J_n} q_n(j)^{\frac{\varepsilon-1}{\varepsilon}} dj\right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

Utility maximization yields demand $q_n(j)$ for each variety and the ideal price index p_n in each location:

$$q_n(j) = \beta_n p_n^{\varepsilon-1} \left(p_n(j)\right)^{-\varepsilon},\ p_n = \left(\int_{J_n} p_n(j)^{1-\varepsilon} dj\right)^{1/(1-\varepsilon)},\ p = \prod_{n=1}^{N} p_n^{\beta_n}.$$

Isoelastic demand means that firms choose a constant markup over marginal cost, $p_n(j) = \varepsilon/(\varepsilon - 1)\, c_n(j)$, where $c_n(j)$ denotes the marginal cost of production of firm j in location n.

Production.—Production combines a primary factor (equipped labor) with an intermediate input, which is sourced from a supplier. There are two stages in the production process: (i) search for suppliers and (ii) sourcing and production. In the first stage, firms in each location n exert bilateral search efforts m_{ni} to search for suppliers from different locations i, maximizing expected profits given their beliefs over flood risk across locations. This search process yields combinations of suppliers and idiosyncratic productivity realizations ("techniques") that firms can use to produce. In the second stage, shocks are realized, and firms choose which of these techniques they will use to maximize profits. We start by describing the second stage, where technique draws and shock realizations are taken as given.

Second Stage: Sourcing and Production.—Search results in the arrival of techniques ϕ, consisting of a supplier s and a match-specific factor-augmenting

productivity z. Each technique describes a Cobb–Douglas production function in equipped labor and intermediate inputs:

$$y_j(\phi) = a_{n(j)} b_{n(j)} \xi_j l_j^{1-\alpha} \big(z(\phi) x_j\big)^{\alpha}, \tag{12}$$

where l_j and x_j are the quantity of equipped labor and intermediate inputs, respectively; z is the match-specific productivity draw; $a_{n(j)}$ is the deterministic time-invariant productivity of location n of firm j; $b_{n(j)}$ is a location-specific productivity shock common to all firms in n, interpreted as coming from floods; and ξ_j is a firm-specific idiosyncratic flood shock. α is the cost share of intermediate inputs.

Suppliers set prices at their marginal cost c_s when they sell to downstream firms—that is, buyers have full bargaining power. Trade is subject to location-specific iceberg costs such that, for each unit to be used as an input in production, $\tau_{n(j)n(s)} \geq 1$ units must be shipped. Denoting the cost of one unit of equipped labor by w, the marginal cost of production using technique ϕ is

$$c_j(\phi) = \frac{1}{a_{n(j)} b_{n(j)} \xi_j} w^{1-\alpha} \left(\tau_{n(j)n(s)} \frac{c_s(\phi)}{z(\phi)} \right)^{\alpha}. \tag{13}$$

Firms choose the technique ϕ, which minimizes marginal cost and maximizes profits.

In this setup, sourcing decisions depend on suppliers' production costs c_s, which in turn depend on their own sourcing decisions, and so on. In order to characterize bilateral sourcing shares in the aggregate equilibrium, we impose two key functional form assumptions. First, following Kortum (1997) and Oberfield (2018), we assume that the distribution of match-specific productivity draws z is such that the number of technique draws where the supplier is in location i and that yield a match-specific productivity z greater than a threshold $\bar{z}$ is Poisson distributed with mean $m_{ni} \bar{z}^{-\zeta}$, where m_{ni} describe bilateral search efforts in the first stage. The parameter ζ governs the tail of the distribution of match-specific productivity draws: Higher ζ implies on average more similar draws, such that a buyer will be more willing to substitute to a different supplier when a supplier experiences an idiosyncratic cost shock. Second, we place a functional form assumption on the distribution of the idiosyncratic flood shocks by assuming that $\xi_j^{\zeta/\alpha}$ follows a positive one-sided stable distribution characterized by its Laplace transform:

$$\mathbb{E}\left(e^{-u\xi_j^{\zeta/\alpha}}\right) = e^{-u^{\beta}}, \beta > \alpha.$$

These assumptions allow us to characterize the distribution of firm production costs in each location:

LEMMA 1: *Conditional on the realization of the aggregate flood shocks b, the cost distribution of firms in location n is Weibull*:

$$P(c_j > c|b) = \exp\left[-\left[\left(a_{n(j)} b_{n(j)}\right)^{\zeta\beta/\alpha} \left(w^{1-\alpha}\right)^{-\zeta\beta/\alpha} \left[\sum_i m_{ni} \tau_{ni}^{-\zeta} \bar{c}_i^{-\zeta} \right]^{\beta} \right] c^{\zeta\beta/\alpha} \right],$$

where location-specific cost indices $\bar{c}_n$ satisfy

$$\bar{c}_n^{-\zeta} = \left(a_{n(j)} b_{n(j)}\right)^{\zeta}\left(w^{1-\alpha}\right)^{-\zeta}\left(\sum_i m_{ni}\tau_{ni}^{-\zeta}\bar{c}_i^{-\zeta}\right)^{\alpha}\Gamma\left(1-\frac{\alpha}{\beta}\right). \tag{14}$$

Sourcing shares follow the common gravity form, with search efforts m_{ni} as bilateral trade flow shifters:

COROLLARY 1: *The expenditure share of location n on inputs from i is*

$$\frac{X_{ni}}{X_n} = \frac{m_{ni}\tau_{ni}^{-\zeta}\bar{c}_i^{-\zeta}}{\sum_{\tilde{n}} m_{n\tilde{n}}\tau_{n\tilde{n}}^{-\zeta}\bar{c}_{\tilde{n}}^{-\zeta}}. \tag{15}$$

First Stage: *Search for Suppliers*.—The first stage of the firms' production problem endogenizes search efforts m_{ni} as the optimal ex ante investments in the face of uncertainty over flood outcomes. Before location-specific flood shocks b_n have been realized, firms in location n have beliefs over the distribution of these shocks described by the information set $\mathcal{I}_n$. Firms are owned by the representative household and maximize profits π discounted by the households' stochastic discount factor λ. They face a resource constraint $\bar{m}$ on the total resources available for supplier search, characterized by a general function g. In the first stage, a firm j in location n chooses search efforts m_{ni} to solve

$$\max_{(m_{ni})_i} \mathbb{E}\left(\lambda\pi_j(m_{n1},\ldots,m_{nN})\,\middle|\,\mathcal{I}_n\right), \tag{16}$$

subject to

$$g(m_{n1},\ldots,m_{nN}) = \bar{m} \quad \text{and} \quad m_{ni} \geq 0 \quad \text{for all } i.$$

We assume that g is such that the solution matrix $\mathbf{m}(\mathcal{I}) = \left(m_{ni}(\mathcal{I}_n)\right)_{ni}$ to this problem is unique.

Equilibrium.—An equilibrium of the economy is a matrix of search efforts $\mathbf{m}(\mathcal{I})$ and cost indices $\bar{c}_n$ such that (i) $\mathbf{m}(\mathcal{I})$ solves the firms' optimal supplier search problem (16); (ii) conditional on the realization of shocks, firms choose techniques to minimize costs and markups to maximize profits; (iii) conditional on the realization of shocks, the representative household maximizes utility; and (iv) goods and labor markets clear.

LEMMA 2: *Let $\alpha > 0$. Then for each realization of the aggregate shocks b_n, an equilibrium exists and is unique.*

Changes in the household price index p or search efforts m in response to flood shocks can be characterized as a function of shocks, elasticities, and observed

preshock equilibrium outcomes. Denoting ratios of a variable in one equilibrium to another by hats ($\hat{x} = x_{post}/x$) yields

$$\log\hat{p}\left(\mathbf{X}, \hat{\mathbf{b}}, \hat{\mathbf{m}}\right) = \boldsymbol{\beta} \cdot \log\hat{\tilde{\mathbf{c}}}\left(\mathbf{X}, \hat{\mathbf{b}}, \hat{\mathbf{m}}\right), \tag{17}$$

where $\mathbf{X} \equiv \left(X_{ni}/X_n\right)_{n,i}$ and $\hat{\tilde{\mathbf{c}}}\left(\mathbf{X}, \hat{\mathbf{b}}, \hat{\mathbf{m}}\right)$ satisfies

$$\hat{\tilde{c}}_n\left(\mathbf{X}, \hat{\mathbf{b}}, \hat{\mathbf{m}}\right) = \hat{b}_n^{-1}\left[\sum_i \frac{X_{ni}}{X_n}\hat{m}_{ni}\hat{\tilde{c}}_i\left(\mathbf{X}, \hat{\mathbf{b}}, \hat{\mathbf{m}}\right)^{-\zeta}\right]^{-\alpha/\zeta}. \tag{18}$$

Dynamics.—The static economy described above is played out in each time period. Time periods are linked by the dynamics of firm beliefs $\mathcal{I}$ over the distribution of floods. Changes in beliefs alter ex ante search decisions for suppliers, $\mathbf{m}\left(\mathcal{I}\right)$, which in turn shift sourcing shares according to the gravity equation (15). We do not impose restrictions on the nature of firm beliefs beyond the requirement that firms in each location update beliefs in the same way, in order to preserve ex ante homogeneity of firms within locations as specified in the model. In our estimation of the model below, we ensure consistency with the empirical results in Section III by defining locations according to flood exposure of a firm's suppliers.

B. *Quantitative Implementation*

This section describes how we take the model in Section IVA to the data in our empirical setting and estimate the aggregate impacts of flood shocks. Section IVC then estimates how postflood adaptation—identified in the model as enduring changes in sourcing behavior driven by updated firm beliefs about flood risk—influences the network's aggregate vulnerability to future flooding.

To estimate the model, we define "locations" as groups of firms. Locations are defined separately for each flood event based on two firm characteristics, which reflect the model's key assumption that all firms in a location n have the same beliefs. First, location definitions reflect whether or not any of a firm's suppliers experience flooding of more than 10 percent of their 2 km buffer in the flood event under consideration. This reflects the evidence in Table 4 that firms whose suppliers are flooded update their beliefs differentially, while the firm's own flood status has no significant impact on firm beliefs.[28] Second, the definition of locations accounts for the district in which the firm is located to permit heterogeneity in beliefs according to the proximity of flooding. Locations are therefore groups of firms that share both characteristics: district and supplier flood status. Sales transactions between firms in each location are aggregated to this level.

Estimation of the model is based on three time periods around each flood event. The flooded period is defined as the six-month period starting when flooding is first recorded, during which the flood reduces TFP in affected locations in line with the dynamics of direct flood impacts in Section II. The six months prior to this are

[28] We restrict attention to firms that report at least ten times in the 13 months of the panel that precede the first flood event in 2012 to ensure that changes in sourcing shares over the six-month periods considered in the estimation are likely to capture true changes in sourcing behavior rather than noise resulting from sporadic reporting.

defined as the preflood period, when no flood is present and $b_n = 1$ for all locations n. The six months subsequent to the flooded period are defined as the postflood period, when the temporary disruptive effects of the flood have subsided and all observed changes in sourcing shares are interpreted as being driven by changes in beliefs. This definition of time periods ensures that, for each flood event, the pre- and postflood periods span the same six months of the calendar year, ameliorating concerns that differences in sourcing behavior may reflect seasonal effects. Technological productivity levels a_n, elasticity parameters, and trade costs τ_{ni} are assumed constant over the three periods of each flood event.

Flood shocks b and ξ are realized independently across flood events. To quantify the consequences of adaptive changes in search decisions, we require a parameterization of location-level productivity shocks induced by flooding. We assume the following form for how these shocks b_n are related to $\overline{ShareFlooded}_n$, the sales-weighted share of firms in n experiencing flooding of more than 10 percent of their 2 km buffer:

$$\log b_n = \eta \log\left(1 + \overline{ShareFlooded}_n\right). \tag{19}$$

We estimate η by substituting this expression into the gravity equation (15), expressed in changes from the preflood period to the flooded period, during which time search decisions m_{ni} are assumed unchanged:

$$\widehat{\left(\frac{X_{nit}}{X_{nt}}\right)} = \exp\left(-\zeta \log \hat{\tilde{c}}_i + \frac{\zeta}{\alpha} \log \hat{\tilde{c}}_n + \frac{\zeta}{\alpha} \log \hat{b}_n\right). \tag{20}$$

Here, variables with hats denote changes from the preflood period to the flooded period, and the changes in cost indices $\hat{\tilde{c}}$ are implicitly defined as a function of shocks $\hat{b}$ and pre-period sourcing shares by

$$\hat{\tilde{c}}_n = \hat{b}_n^{-1} \left[\sum_i \frac{X_{ni}}{X_n} \hat{\tilde{c}}_i^{-\zeta}\right]^{-\alpha/\zeta}. \tag{21}$$

Estimation of equation (20) using observed sourcing shares, together with values for the parameters α and ζ, can be used to obtain an estimate of η.[29] We set the input share α equal to the average annual share of reported purchases to sales, which is 0.77.[30] The trade elasticity ζ is calibrated to be 4 following Simonovska and Waugh (2014).

This estimation yields an estimate of $\eta = -0.42$, which implies that a location where all firms saw flooding of more than 10 percent of their 2 km buffer would experience a 25 percent reduction in TFP. This estimate of η, together with the observed share of firms in each location n experiencing flooding of more than

[29] We ensure that all observations of shares are finite and reduce the influence of outliers by estimating equation (20) on the set of *cell pair* × *events* (n, i, t^*) where, prior to the flood, the purchases of n from i account for at least half a percent of n's purchases, and winsorizing the change in purchase shares at the ninety-ninth percentile.

[30] In this calculation we ignore firms that report fewer than three times in a year and firms that have purchase-to-sales ratios exceeding 3. This value is larger than most materials shares reported in the literature because purchases from firms can also include capital.

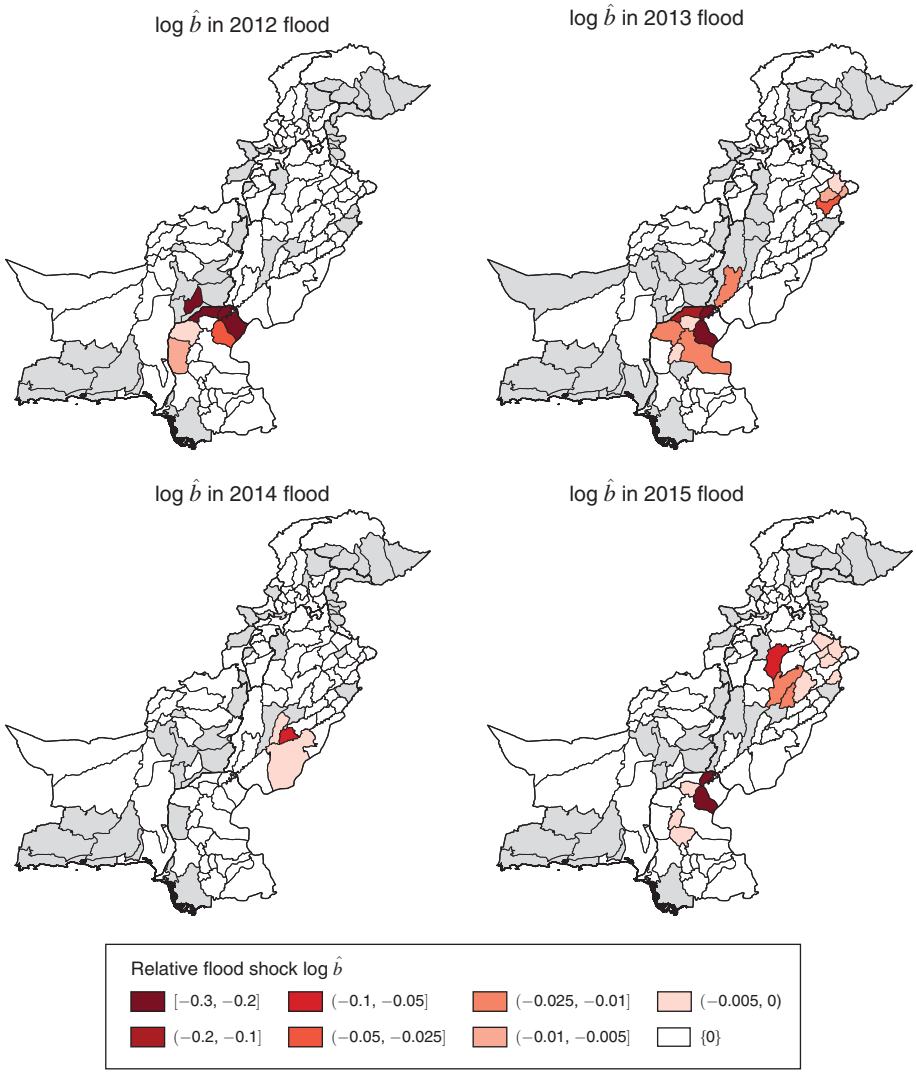


FIGURE 9. ESTIMATED PRODUCTIVITY SHOCKS FROM FLOODS IN SAMPLE

Notes: The figure maps the productivity shocks log $\hat{b}$ estimated from equation (19) as described in this section. We obtain log $\hat{b}$ at the district level by taking the average across both locations in the district weighted by preflood period sales to households.

10 percent of their 2 km buffer, yields the productivity cost b_n of each flood in each location from equation (19). These values are aggregated to the district level and mapped for each flood event in our sample in Figure 9. While the majority of locations do not experience direct reductions in TFP in a given flood event, the maximum decrease across locations ranges from 20 percent to 29 percent.

The impact of flooding on firm costs extends beyond direct location-level TFP reductions via the general equilibrium effects of flood exposure of firms' suppliers (as well as that of their suppliers' suppliers, and so on). The total increase in firm costs in each location accounting for such effects is calculated from equation (21) and plotted for the flood events in our sample in Figure 10.

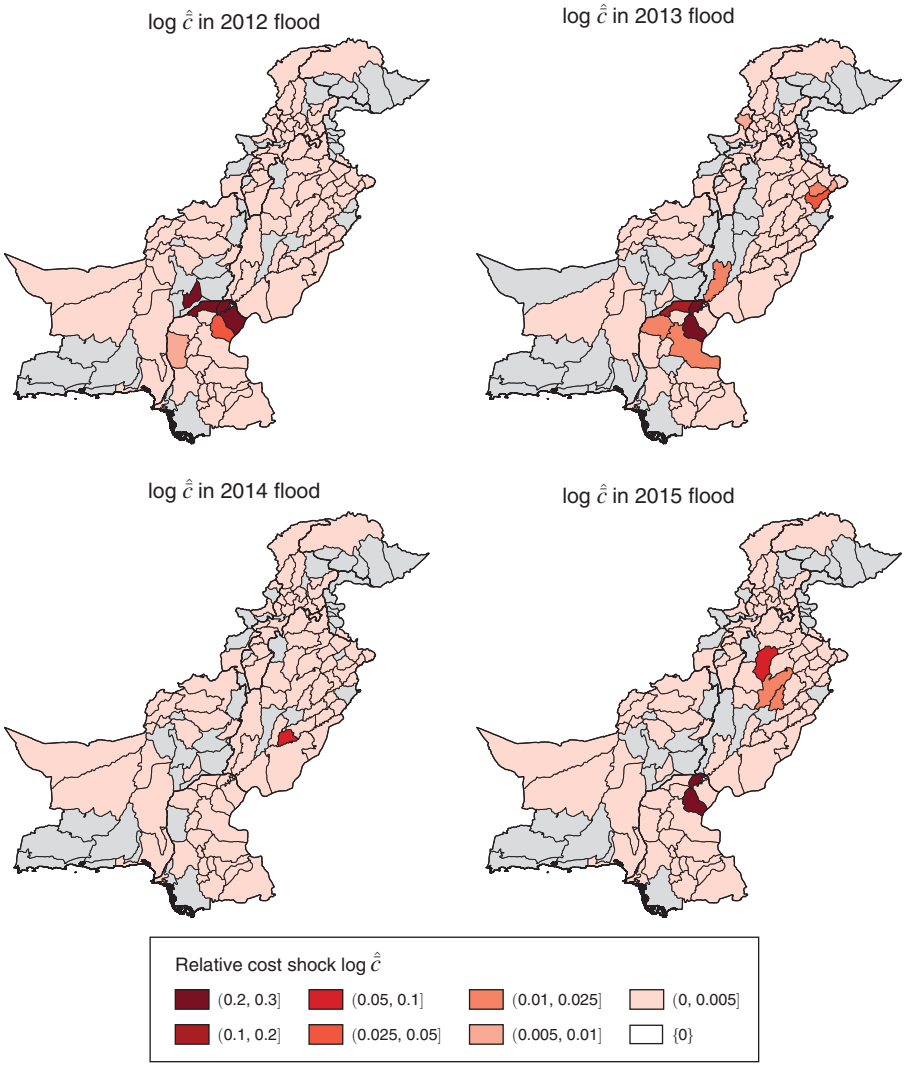


FIGURE 10. ESTIMATED INCREASES IN FIRM COSTS ACROSS LOCATIONS FROM FLOODS IN SAMPLE

Notes: The figure displays the increases in firm costs $\log \hat{\hat{c}}$ estimated from equation (21) as described in this section. We obtain $\log \hat{\hat{c}}$ at the district level by taking the average across both locations within the district weighted by preflood period sales to households.

The economy-wide impact of each flood event can be estimated as the increase in the households' cost index p. This is calculated by aggregating location-level cost increases using equation (17), where the final demand shares β_n are calibrated to the share of each location's sales to out-of-network buyers. The results yield estimated increases in the households' cost index as a result of the 2012, 2013, 2014, and 2015 floods of 0.05 percent, 0.30 percent, 0.06 percent, and 0.16 percent, respectively.[31]

[31] To benchmark the magnitude of these estimates, estimated total annual direct economic losses from all categories of natural disasters in Pakistan between 2000 and 2013 averaged 1.16 percent of the national GDP, including substantial losses from the severe 2010 floods and 2005 earthquake (ADB 2021).

C. *Estimating the Aggregate Impacts of Adaptation*

In this section, we use the estimates from Section IVB to quantify the impacts of observed postflood adaptation for the aggregate damages from subsequent floods. We quantify this by calculating the costs of a subsequent flood with and without incorporating the adaptive sourcing changes induced by a prior flood. These changes in sourcing decisions reflect the aggregate effect of upstream firms exiting or moving away from affected areas, and buyers choosing to purchase from less risk-prone areas, thereby capturing all relevant adaptation margins highlighted in Section III.

Equation (21) yields cost changes $\hat{\tilde{c}}$ induced by flooding using estimates of the flood shock $\hat{b}$ and the baseline sourcing shares X_{ni}/X_n. To simulate the role that adaptation following a prior flood plays in changing the costs of a subsequent flood, we solve this equation once under the sourcing shares that prevailed before adaptation to the prior flood, and once under the postadaptation sourcing shares, and compare the resulting cost increases induced by the subsequent flood.

We consider the example of adaptive changes in sourcing shares undertaken following the observed flood in 2012 and estimate the impacts of these changes in shaping the damages imposed by subsequent floods in the sample. This reveals that the 2013 flood would have resulted in a 5 percent higher increase in the household price index in the absence of adaptation following the 2012 flood, as captured by changes in the pre- to post-2012 flood sourcing shares.[32] Similarly, damages from the 2015 flood would have been 1 percent higher. While this suggests that adaptation in the aftermath of the 2012 flood helped in ameliorating damages from the subsequent floods in 2013 and 2015, this need not be the case for all future flood shocks. In contrast, adaptation following the 2012 floods relocated sourcing activity *toward* areas that would subsequently be affected by the 2014 floods, and as a result damages from the 2014 floods would have been 3.5 percent *lower* without adaptation following the 2012 flood.[33]

To understand these patterns, consider the spatial distribution of flood impacts shown in Figures 9 and 10. The 2012 and 2013 floods can be seen to affect very similar areas. Consistent with this, sourcing share changes undertaken following the 2012 floods shift activity away from regions that will face flood exposure in 2013 and as such attenuate damages from a 2013 flood scenario. Conversely, the 2014 floods afflict quite different regions, such that adaptive changes in sourcing shares following the 2012 floods do not necessarily move activity from areas that will be exposed in 2014, and in fact are associated with *higher* damages in this case. The 2015 flood is intermediate between these two cases, affecting both areas that were and were not flooded in 2012, so that adaptation following the 2012 floods had more muted protective impacts for the 2015 floods. As such, the pattern of gains and losses from adaptation will depend intuitively on the spatial correlation between areas that firms shift their sourcing toward or away from following one flood event, and areas adversely affected by a subsequent event.

[32] Aggregation weights locations by their share of out-of-network sales in the period preceding the adaptation year.

[33] Noisy measurement of these sourcing shares attenuates the impacts of these adaptive changes. Estimates obtained using manufacturing firms only—for which sourcing shares tend to be more stable—are a few percentage points larger but of the same order of magnitude.

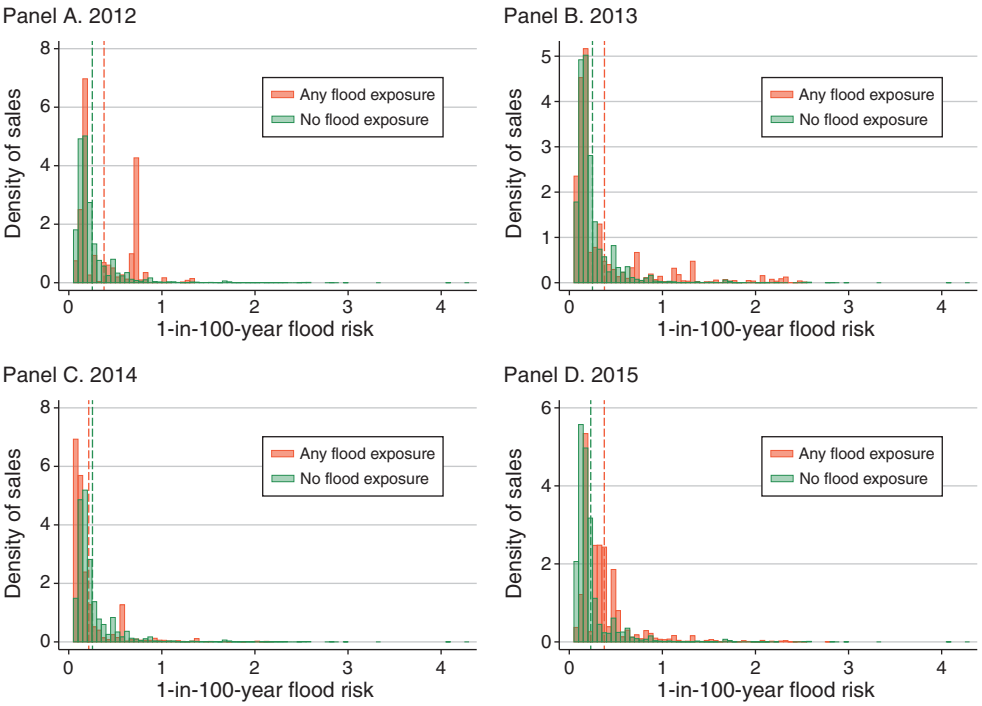


FIGURE 11. DENSITY OF SALES ACROSS 1-IN-100-YEAR FLOOD RISK BY FLOOD EXPOSURE

Notes: The figure plots the density of sales by flood exposure in different flood events. Dashed lines indicate sales-weighted average flood risk. Sales refer to total declared sales between July 2011 and August 2012. Flood risk is defined as the expected flood depth in meters for a 1-in-100-year return period. Flood exposure indicates whether the firm had a positive share of its buffer flooded during the flood event.

The economy-wide aggregate figures mask substantial heterogeneity in the effects of adaptation across regions. While overall damages in a 2013 flood scenario would have been 5 percent higher in the absence of adaptation following the 2012 floods, flood damages would have been at least twice as large in four regions. The benefits of adaptation are also widely felt in this case, with 94 percent of locations benefiting from the prior adaptation. Conversely, aggregate damages in a 2015 flood scenario would have been 1 percent higher without adaptation following the 2012 floods, with only 68 percent of locations seeing benefits from adaptation.

The heterogeneous impacts of adaptation following the 2012 flood across subsequent floods in the sample raise a question as to the conditions under which postflood adaptation to reduce flood *risk* may be expected to help or hurt in the face of individual future disasters. Some guidance on this question can come from considering how far the locations flooded in each of the floods in our sample coincide with regions of high flood risk in the Fathom data. Intuitively, for the majority of floods, the flood risk distribution among flooded firms is rightward shifted relative to that of their nonflooded counterparts, as shown in panels A, B, and D of Figure 11. Flood events are, however, stochastic, and may also hit lower flood risk areas. Indeed, panel C of Figure 11 shows that firms affected by the 2014 flood are on average *less* flood prone than those that are not. This is exactly the flood for which adaptation following the

2012 flood worsened outcomes—and is the outlier in other pairwise comparisons of the impact of postflood adaptation on the damages from subsequent flood events (see Supplemental Appendix Table A.10). This suggests that postflood adaptation may be more likely to reduce damages from future floods on average where floods affect flood-prone regions, but that the converse may occur for idiosyncratic events that affect areas that are not especially flood prone.

D. *Evaluating the Potential for Adaptation*

The analysis above reveals that adaptive sourcing adjustments can yield benefits from reduced future flood exposure. In a final structural exercise, we evaluate the economy's remaining potential to reduce the impact of climate-related shocks by adjusting supply chains. Concretely, we use the model to calculate how many locations could adjust their input sourcing shares by a small amount, $d\log\left(X_{ni}/X_n\right)$, such that there is a decrease in the variance of cost changes induced by flood shocks, $var_{\hat{b}}\left(\hat{c}_n\right)$, while ensuring that expected costs $\mathbb{E}\left(c_n\right)$ do not increase. This gives a measure of the potential for locations to shift toward suppliers that result in lower exposure to cost variability from flooding, but which do not increase expected costs on average.

For this exercise, we calculate the variance and expectation over the joint distribution of flood shocks across locations, parameterized using a longer time series of historical flood data in Pakistan as described in Supplemental Appendix E. We implement this exercise[34] at the level of the spatial aggregates used for the 2015 flood event (as described in Section IVB) in order to estimate the remaining potential for adaptation at the end of our sample period; similar but slightly larger results obtain using the baseline sourcing shares from 2012.

The results of these simulations indicate that 80 percent of locations could reduce cost variability from flooding by altering their supply chain exposure without increasing expected costs on average.[35] This significant scope to reduce the volatility associated with climate shocks without incurring large costs highlights the opportunity for policies that support firms in realizing this potential for cost-effective adaptation.

V. Conclusion

The impact of climate change depends crucially on the ability of economic agents and systems to adapt to its effects. The results of this paper suggest a consequential role for adaptation by firms. We demonstrate that natural disaster events—a key manifestation of climate change—can also be important in influencing its impacts by inducing firm-level adaptation in their production and network linkage decisions. Floods result in only temporary disruption to production and transportation links, but prompt persistent shifts in firm location, supplier, and route choice that reduce firms' vulnerability to the recurrence of such events in the future. These responses

[34] See Supplemental Appendix D.2 for the precise mathematical definition of the problem and the computational strategy.

[35] In our model, firms may of course not attempt to minimize the variance of cost increases, but may target other moments that depend on their risk aversion and stochastic discount factor. Firms may also not realize these potential gains if they do not know the true distribution of flood shocks.

are enduring, consistent with flood events causing firms to update their beliefs about underlying flood risk.

The interdependent nature of firm supply chains and the central role of vertical linkages in adaptation suggest that firm adjustments may have important general equilibrium implications for other firms and the resilience of the aggregate firm network. To quantify this, we develop a spatial equilibrium model of firm production and sourcing decisions under imperfect information about flood risk. Flood shock realizations percolate through the spatial production network and raise the cost of production. The model allows us to evaluate the impact of flood realizations both with and without adaptive supplier sourcing changes. Estimating these effects reveals that firms' adaptive behavior following floods observed in our sample has quantitatively meaningful implications for the damages imposed by future flood events on the aggregate economy.

These findings open up a research agenda on how far private adaptation may shape future climate damages. First, as statistical models of the distribution of extreme weather events under different climate change scenarios evolve, the frameworks developed here can be used to evaluate the dynamic impact of adaptation events across the full distribution of future flood risk. Second, adaptation is triggered by a multitude of events, not just disaster episodes as studied here, raising important policy questions about whether complementary approaches might also induce adaptation—for instance, could providing accurate information to firms on flood risk be sufficient to induce meaningful adaptation, or do firms only respond to costly flood experience? Third, adaptive responses by firms likely interact with those of households and the public sector, and a better understanding of these interdependencies will be required to estimate aggregate impacts. Finally, our results relating to firms' imperfect information about climate risk highlight the need for further analysis of their expectations about long-range climate trajectories, including to better understand whether belief updating and adaptive behaviors attenuate over time as major flood events become more common (Belasen and Polachek 2008). Given sharply deteriorating projections of natural disaster incidence as climate change proceeds, understanding the nature, determinants, and implications of firm adaptation will be crucial in projecting and responding to future damages.

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